OWL or SHACL?

A Beginner's Guide to Making the Right Choice

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KGC The Knowledge Graph Conference

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The Ultimate Showdown: Game 1

Created using ChatGPT+Gemini

What we are learning today

- This will be a hands-on, example-based tutorial
  - Together, we will make a choice of OWL or SHACL (or both) to solve problems and understand the implications of the choice
- Main Focus: Comparing the strengths and weaknesses of OWL and SHACL
- We will talk about
  - Open World and Closed World assumption
  - Reasoning
  - Classification
  - Transitivity and Property chains
  - And more…..
- This will NOT be an in-depth tutorial of everything OWL or everything SHACL… if only we had more time :)

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Some Logistics

- Download Protégé Desktop (NOT WebProtégé): https://protege.stanford.edu/
- Open SHACL playground: https://shacl-playground.zazuko.com/
- Open workshop page for all examples and code: https://tinyurl.com/owlshaclkgc2026

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When you see these icons

Go to Protégé

- For modeling in owl
- Adding data instances
- Using a reasoner
- Running SPARQL

Go to SHACL playground

- For writing SHACL shapes
- For validating data graphs against SHACL shapes

Follow the instructions

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A Little Background

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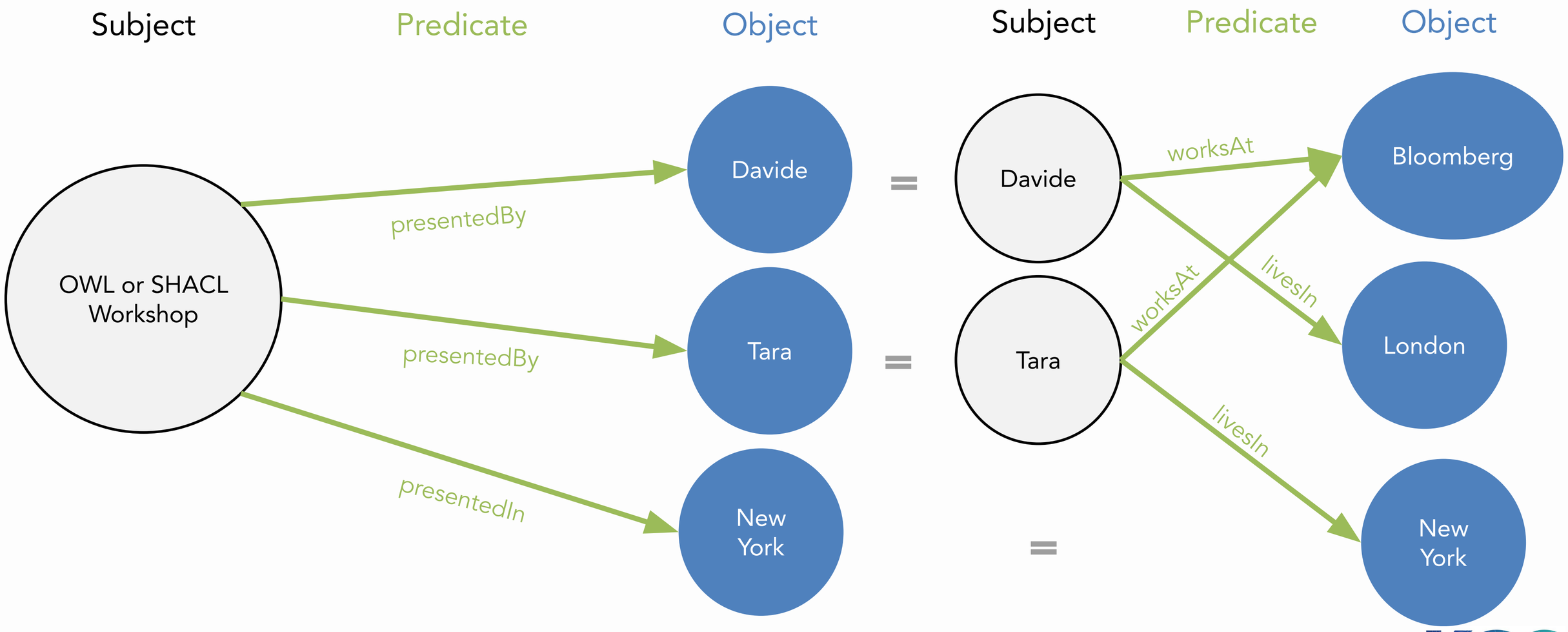
The World in Triples



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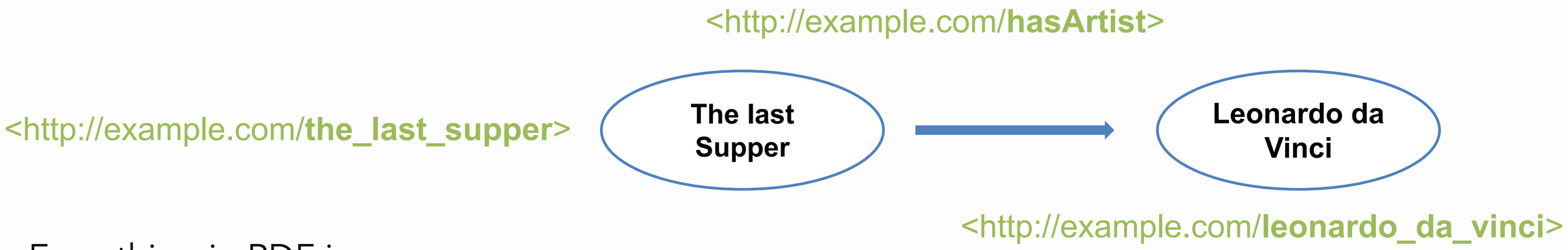
RDF (Resource Description Framework)

- W3C de facto standard for data interchange that is used for representing highly interconnected data
- Simple triple-based model: <subject> <predicate> <object>



- Everything in RDF is a resource
- Every resource has a unique identifier (URI or IRI)
- No schema

Things, not strings!

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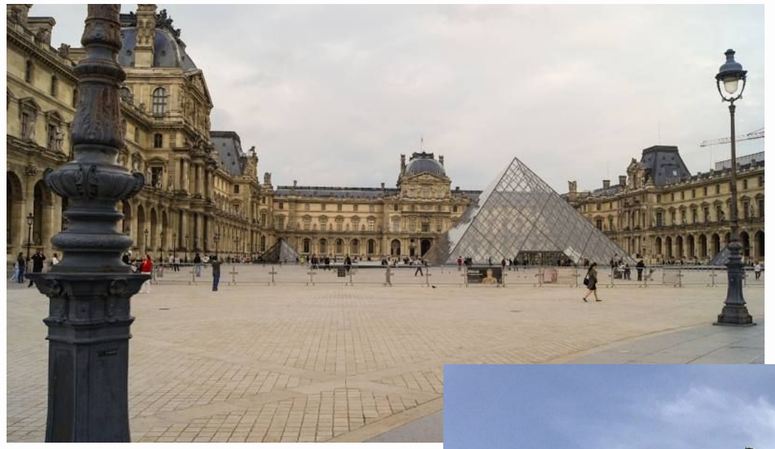
A day at the museum …



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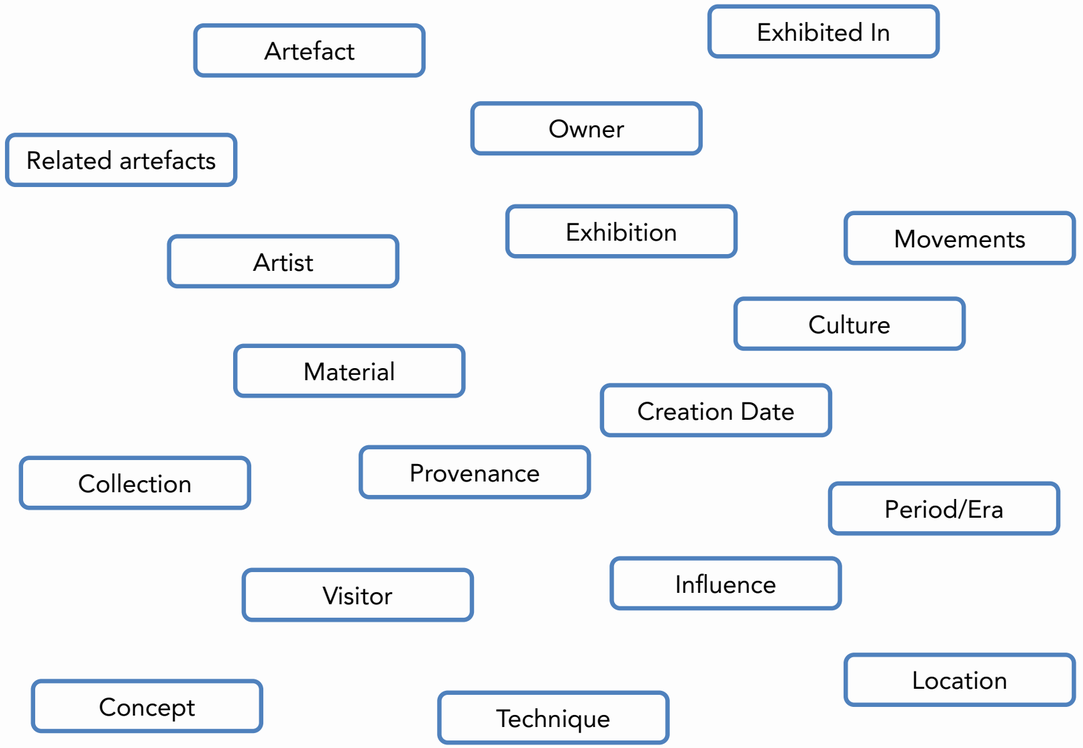
Walking through an exhibition …. A million questions!

- What artefacts do you have?
- All the information about them in terms of artist, date of creation, materials, etc.?
- When you create the exhibitions how do you decide what artefacts go together?
- Can you know which city or county an artefact is being exhibited?
- Based on the art movement era of the piece can you tell the artists of those eras?
- Is it possible for you to know whether a copy of an original art pieces you have exists somewhere else?
- Which piece was influenced by whom?

How do they organize their data?



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ROUND 1:
Simple queries

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Let’s create the ontology!

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How do you decide on the design of the ontology model?

Start with the competency questions…

- What artefacts are attributed to a specific artist?
- What artefacts are associated with a specific historical movement?
- What artefacts are attributed to a specific artist and associated with a specific historical movement?

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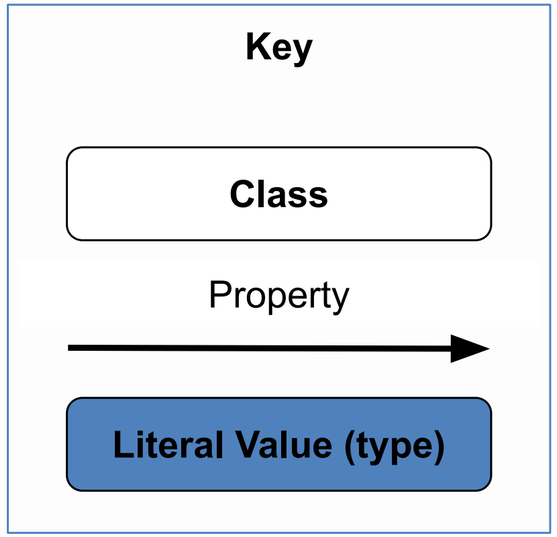
The Simple Ontology - Let's code that in OWL

- Open model_00.ttl
- Add the above classes and properties

  Or
- Open model_01.ttl



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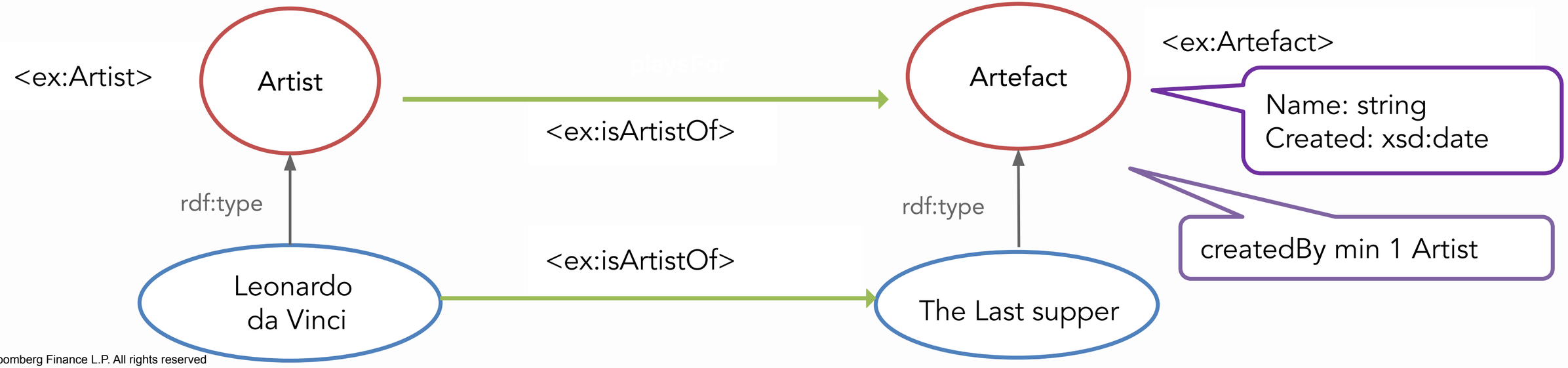
OWL (Web Ontology Language)

- An RDF-based de facto standard for ontology development
- Main components include:
  - Classes: Define concepts in a domain
  - Properties:
    - Object properties: Define relationships between concepts
    - Datatype properties: Define relationships between a concept and a literal
  - Individuals: Instances of classes
  - Restrictions: Allow definition of cardinality restrictions, as well as existential & universal quantifications (some rules can be defined in the context of a restriction)



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Let’s add the data

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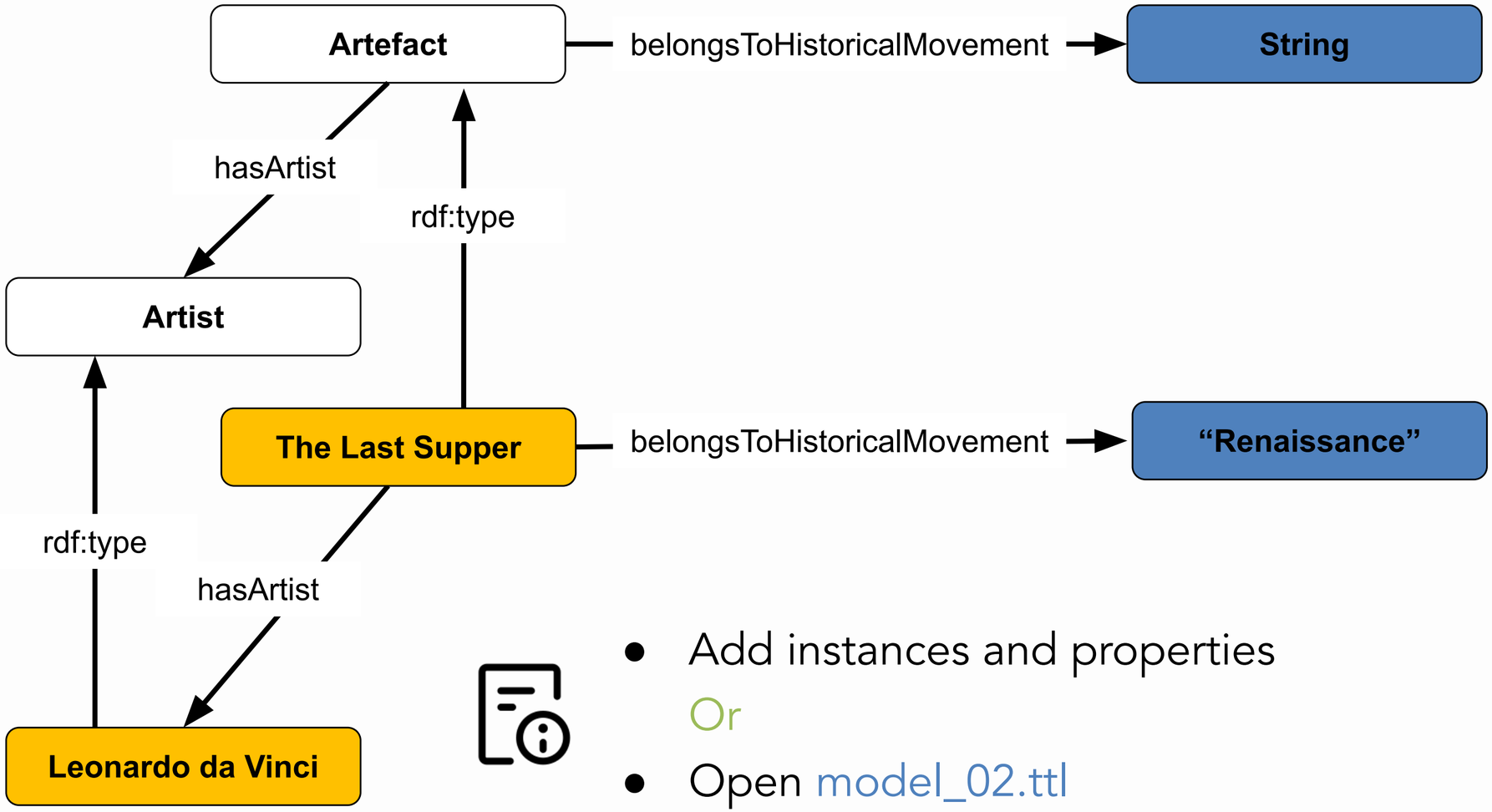
The Knowledge Graph



- Add instances and properties

  Or

- Open model_02.ttl

Fun Fact:

In 2007, an Italian musician claimed to have found musical notes in *The Last Supper* that result in a tuneful 40-second musical composition. The notes, which are ‘hidden’ in the bread rolls and hands of the apostles in the painting, is read from right to left, following da Vinci's own writing style.

Source:https://www.livescience.com/2039-da-vinci-musical-code-supper.html

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What is a Knowledge Graph?

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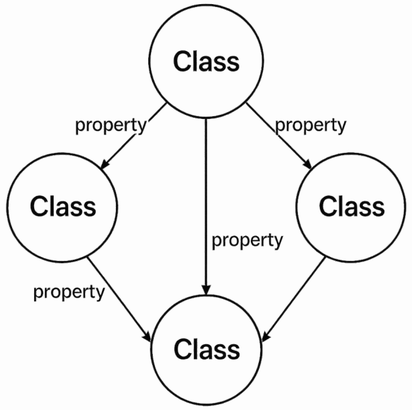
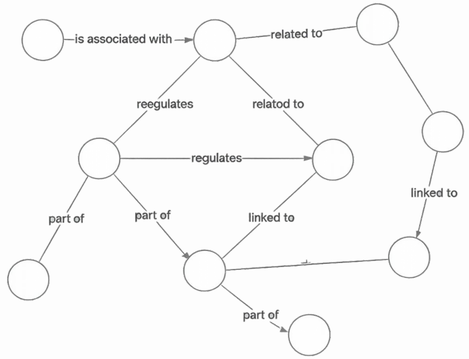
The two layers of the Knowledge Graph



Semantic Layer

Defines what things mean and how they relate to each other in a machine-understandable format



Data Layer

Real-world facts and observations mapped onto the semantic structure/layer

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What is a Knowledge Graph?

"A knowledge graph is the DNA of information —
It encodes not just facts, but the structure, relationships, and rules that bring raw data to life and allow it to grow, adapt, and evolve."

- Tara Raafat

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Let’s test the queries

1. What artefacts are attributed to Leonardo da Vinci?

```
SELECT ?x
WHERE {
   ?x a ex:Artefact ;
     ex:hasArtist ex:leonardo_da_vinci .
}
```

2. What artefacts are associated with the Renaissance movement?

```
SELECT ?x
WHERE {
   ?x a ex:Artefact ;
     ex:belongsToHistoricalMovement "Renaissance"^^xsd:string
.
}
```

3. What artefacts are attributed to Leonardo da Vinci and associated with the Renaissance movement?

```
SELECT ?x
WHERE {
   ?x a ex:Artefact ;
     ex:hasArtist ex:leonardo_da_vinci ;
     ex:belongsToHistoricalMovement "Renaissance"^^xsd:string
. }
```

- Open model_02.ttl
- Open ‘SPARQL” tab in the website
- Run the queries in Section 1

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Can we answer those questions using SHACL?

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What is SHACL?

SHACL shapes are required to validate RDF data against specific structural and value constraints.

We need to define the exact shape of our data, including the required properties and any constraints on their values.

In SHACL, a shape is composed of targets and constraints.

- Targets identify the nodes in the data graph that the shape applies to.
- Constraints specify the rules that those nodes must satisfy in order to be considered valid.

Declare constraints directly on a node

Targets

Node shape

```
ex:ArtefactShape a sh:NodeShape ;
  sh:targetClass ex:Artefact ;

  sh:property [
    sh:path ex:hasEndCreationDate ;
    sh:datatype xsd:integer ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
  ] ;
```

Property shape

Constraints

Declare constraints on the property that is connected to the node through a path (the sequence of edges through which a property is connected to a node)

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Targets in SHACL

sh:targetClass

```
ex:ArtefactShape a sh:NodeShape ;
  sh:targetClass ex:Artefact ;

  sh:property [
    sh:path ex:belongsToHistoricalMovement ;
  ] ;
```

sh:targetNode

```
ex:ArtistShape a sh:NodeShape ;
sh:targetNode ex:Leonardo_Davinci;
sh:nodeKind sh:IRI;
sh:property [sh:path ex:hasDateOfBirth ;
                     sh:range xsd:string ].
```

sh:targetSubjectOf

```
ex:SculptureShape a sh:NodeShape ;
  sh:targetSubjectOf ex:hasArtist ;
  sh:property [
    sh:path ex:hasEndCreationDate ;
    sh:datatype xsd:integer ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
  ] ;
```

sh:targetObjectOf

```
ex:SculptureShape a sh:NodeShape ;
  sh:ObjectOf ex:hasArtist ;
  sh:property [
    sh:path ex:hasName ;
    sh:datatype xsd:string ;
    sh:minCount 1 ;
    sh:maxCount 1 ; ] ;
```

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SHACL to answer simple queries

Yes!

Shape 1 in SHACL tab of website

```
ex:ArtefactShape a sh:NodeShape ;
  sh:targetClass ex:Artefact ;

  sh:property [
    sh:path ex:belongsToHistoricalMovement ;
    sh:datatype xsd:string ;
  ] ;

  sh:property [
    sh:path ex:hasArtist ;
    sh:class ex:Artist ;
  ] .
```

Since you are querying the data, same SPARQL will provide the same answer.

If you want to query the model then you have to write different SPARQL queries one for the OWL construct and one for the SHACL constructs.

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX sh: <http://www.w3.org/ns/shacl#>
SELECT *
WHERE {
    ?s a sh:NodeShape .
}
```

However, if you want to use the same SPARQL query for both, you'll need a tool that translates SHACL to OWL.

But what if you want to do more?

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What if data is missing?

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ROUND 2:
Dealing with incomplete information

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What about missing (incomplete) information?

Artworks are acquired without complete information at the time of acquisition.

- How do museum systems typically handle missing data?
- Does the absence of certain metadata trigger errors?
- Should incomplete records be treated as errors?

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Let’s add more Data

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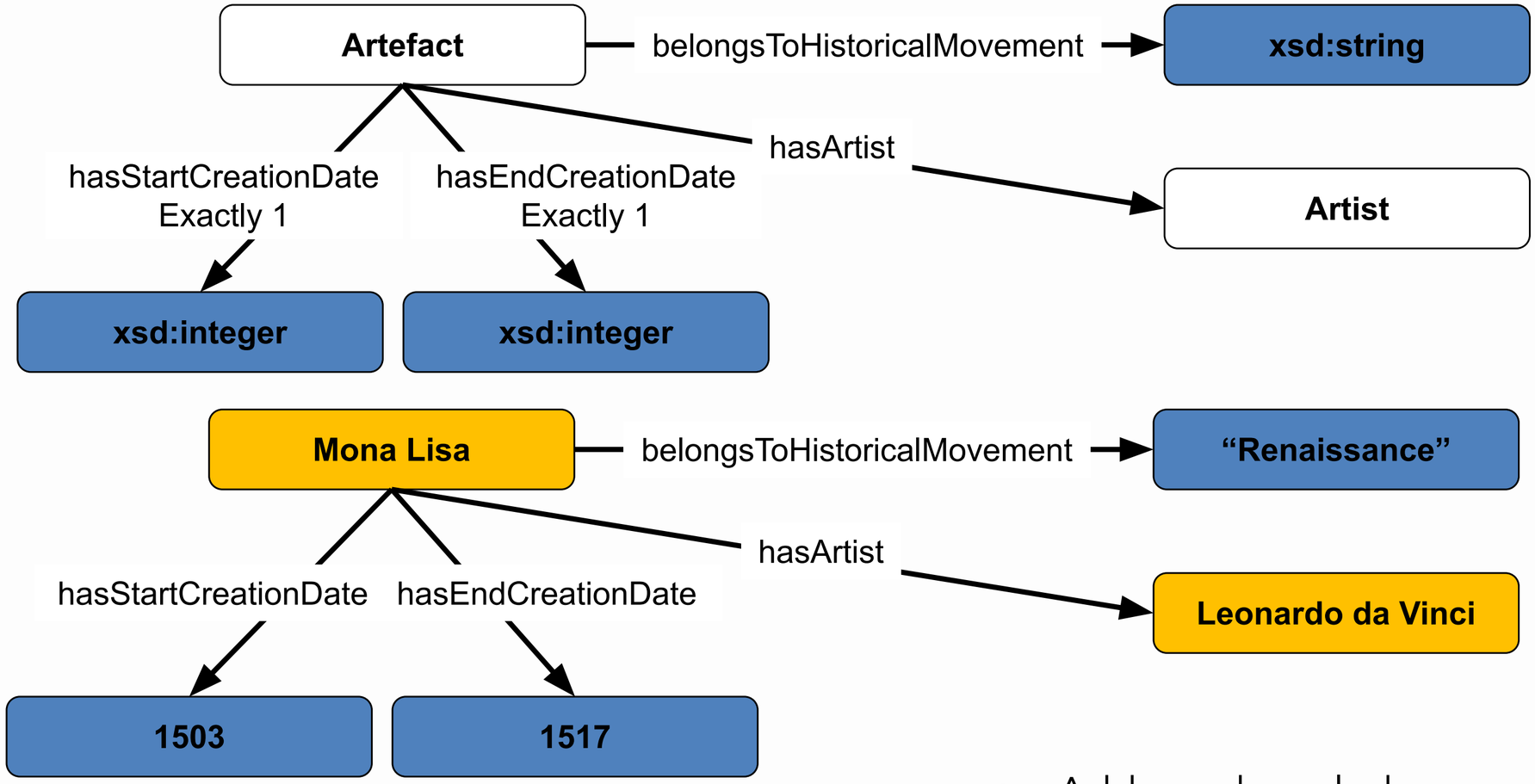
The Knowledge Graph extended



- Add new knowledge
  Or
- Open model_03.ttl

Fun Fact:
Leonardo da Vinci's *Mona Lisa* took 14 years for him to complete, starting in 1503 and continuing until nearly his death in 1517; reportedly, he continuously refined the painting throughout this period.

Source: Wikipedia

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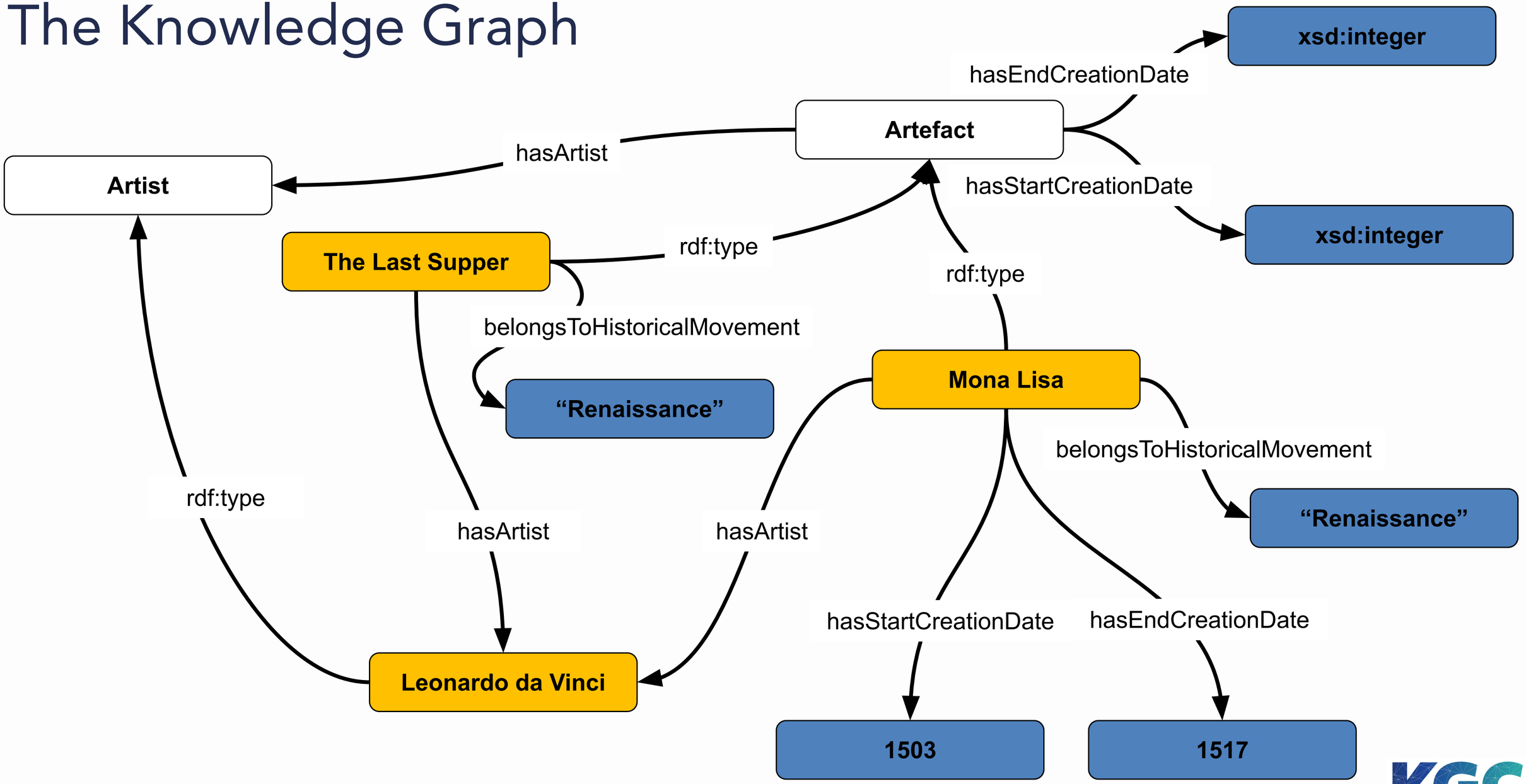
The Knowledge Graph



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Would OWL restrictions enforce data's existence & cause an error?

Artefact := **hasStartCreationdate** **exactly** **1** **xsd:integer**

Artefact := **hasEndCreationdate** **exactly** **1** **xsd:integer**

- Load model_03.ttl
- Run reasoner

The reasoner doesn't throw an error.

Why? Because OWL operates on an Open World Assumption (OWA) and assumes that the absence of information does not imply its absence in reality.

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Open World Assumption (OWA)

If certain information does not exist in the data, it cannot be assumed to be false, it's simply incomplete.

In such cases, an OWL reasoner does not return an error; it simply does not infer anything additional from the missing information.

When is the OWA useful?

The OWA is particularly valuable in scenarios where not all knowledge is known or available at a given time. In such contexts, the use of an ontology under OWA supports ongoing discovery by remaining flexible, adaptable, and collaborative.

New information can be added over time without invalidating existing assertions, enabling systems to grow and evolve without requiring full knowledge upfront.

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What if you wanted to ensure the data existed at the time of acquisition?

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ROUND 3: Validation

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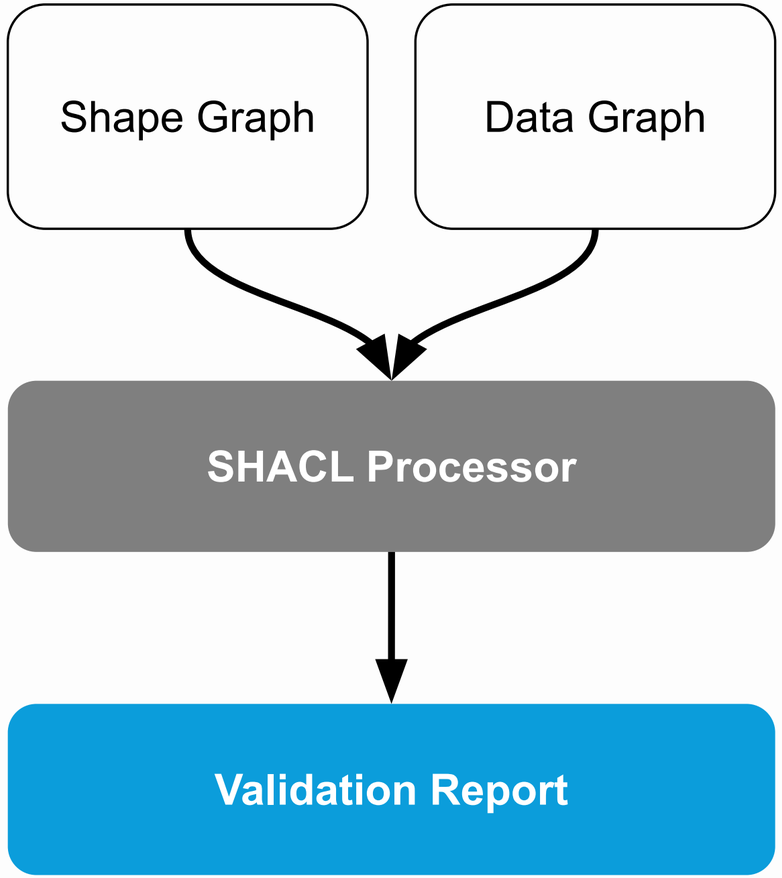
Let's test SHACL




```
ex:ArtefactShape a sh:NodeShape ;
  sh:targetClass ex:Artefact ;

  sh:property [
    sh:path ex:belongsToHistoricalMovement ;
    sh:datatype xsd:string ;
  ] ;

  sh:property [
    sh:path ex:hasArtist ;
    sh:class ex:Artist ;
  ] ;

  sh:property [
    sh:path ex:hasEndCreationDate ;
    sh:datatype xsd:integer ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
  ] ;

  sh:property [
    sh:path ex:hasStartCreationDate ;
    sh:datatype xsd:integer ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
  ] .
```

Shape 2

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Validity Conditions

```
sh:property [
  sh:path ex:hasEndCreationDate ;
  sh:datatype xsd:integer ;
  sh:minCount 1 ;
  sh:maxCount 1 ;
] ;

sh:property [
  sh:path ex:hasStartCreationDate ;
  sh:datatype xsd:integer ;
  sh:minCount 1 ;
  sh:maxCount 1 ;
] .
```

The validation shows error.

Why? Because SHACL operates on an Closed World Assumption (CWA) and assumes that the absence of information is an error in reality.

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Closed World Assumption (CWA)

- The system assumes it has all the information it needs about its domain.
- Anything not explicitly known or stated as true is assumed to be false => Missing information is assumed to be false, the data graph is flagged as invalid.
- If a data node does not meet the defined restrictions, such as a wrong value type for a property, it results in a validation error.

When is the CWA useful?

The CWA is particularly valuable in scenarios where data integrity is crucial and error reports need to be generated if information is missing or invalid.

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Recap so far…

- Open World: Missing information does not cause an error OWL
- Closed World: Missing information causes and error SHACL
  - Data Integrity: SHACL
  - Data Validation: SHACL

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ROUND 4: Classification

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What about Classification?

Museums regularly acquire new artefacts and classify them according to internal categorization systems.

Can this classification be automated? For instance, could a system infer whether an artefact is a painting or a sculpture based on metadata available at the time of acquisition?

```
ex:david
        ex:belongsToHistoricalMovement “Renaissance” ;
        ex:hasArtist ex:michelangelo ;
        ex:hasEndCreationDate 1504 ;
        ex:hasStartCreationDate 1501 ;
        ex:hasSculptureMaterial “Marble” ;
        rdfs:label "David" .
```

- On model_03.ttl
- Add start and end creation dates for ‘The Last Supper’ (1494-1498)
- Add the data about David on the right

Or

- Open model_04.ttl

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Let’s extend our ontology

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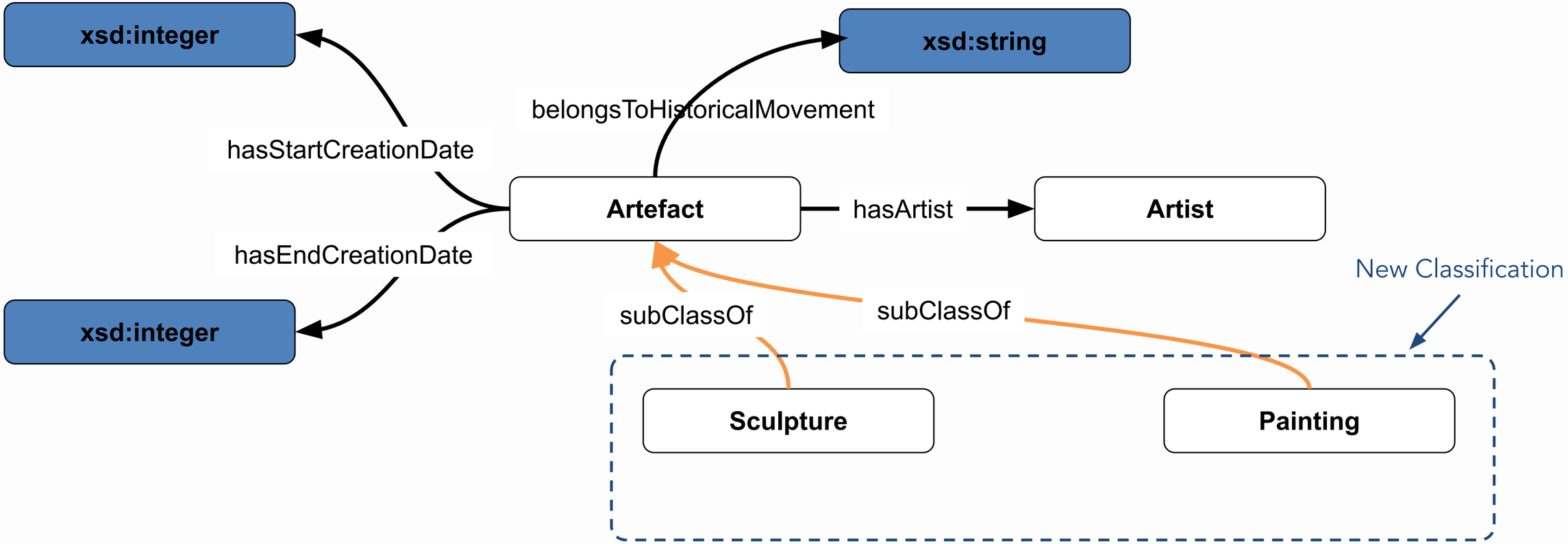
Ontology extension



- Add two new classes Sculpture and Painting
  Or
- Open model_05.ttl

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Classification Query

Using the new model and the data we had about David sculpture;
Can we query the data to see what's sculptures we have?

Let's query the data for Sculpture (SPARQL query 2.1):

```
SELECT ?x

WHERE {
    ?x a ex:Sculpture .
}
```

Nothing!
Why? There is neither a definition for the class sculpture nor a definition of the property hasSculptureMaterial.

Fun Facts:
The statue was carved from a single block of marble.

David's right hand is disproportionately large compared to his body because in the Middle Ages, David was said to be "manu fortis", meaning strong of hand.

There are at least 30 full-size replicas of this statue around the world.

Source:
https://www.contexttravel.com/blog/articles/ten-facts-about-the-statue-of-david

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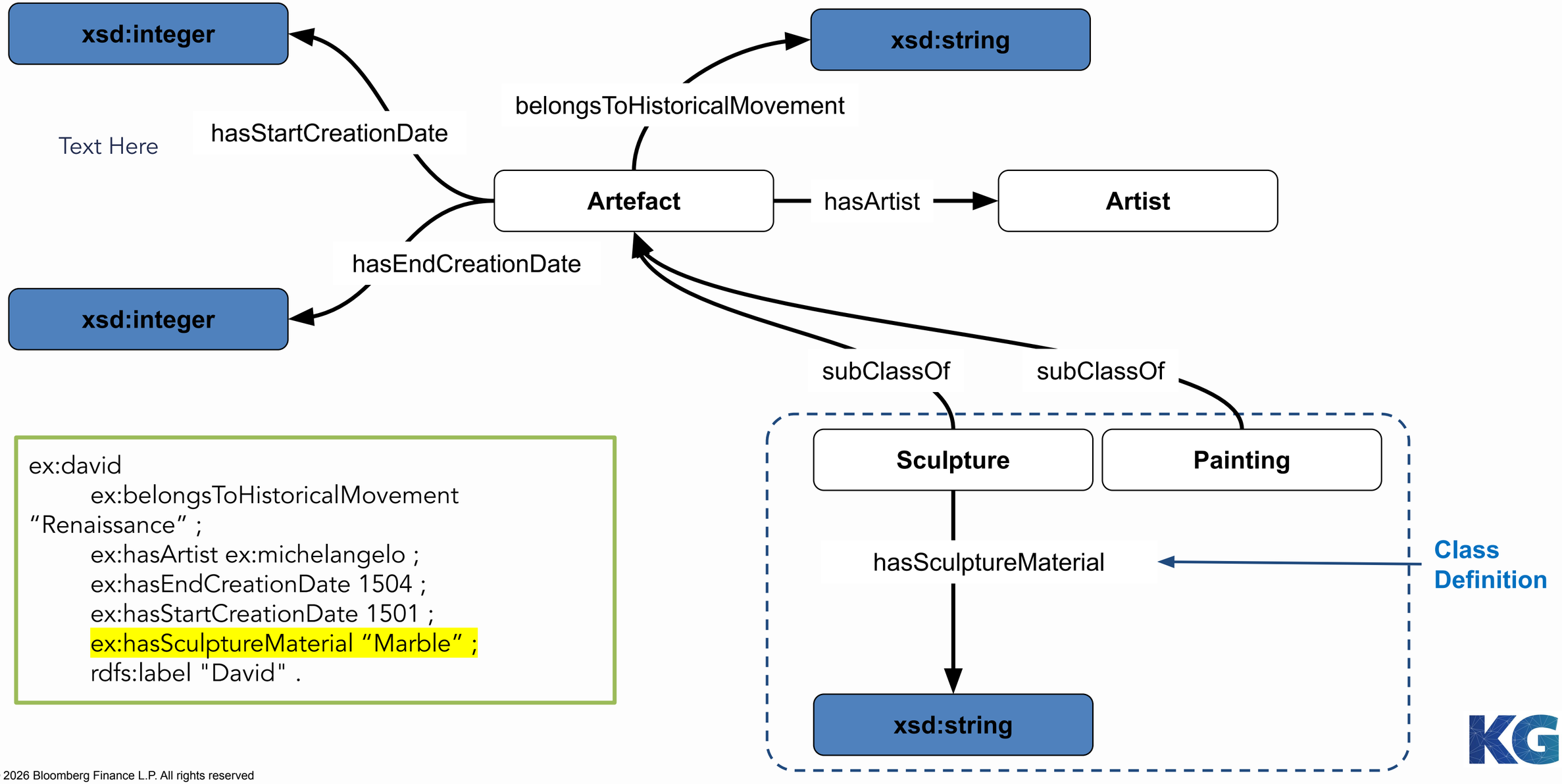
Class Definition




```
ex:david
        ex:belongsToHistoricalMovement
"Renaissance" ;
        ex:hasArtist ex:michelangelo ;
        ex:hasEndCreationDate 1504 ;
        ex:hasStartCreationDate 1501 ;
        ex:hasSculptureMaterial "Marble" ;
        rdfs:label "David" .
```

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Classification by Class Definition

- Add the class definition
  Or
- load model_06.ttl
- Run the reasoner
- Add the changes suggested by the reasoner
  Or
- upload model_06_inferred.ttl
- Run the query 2.1 again

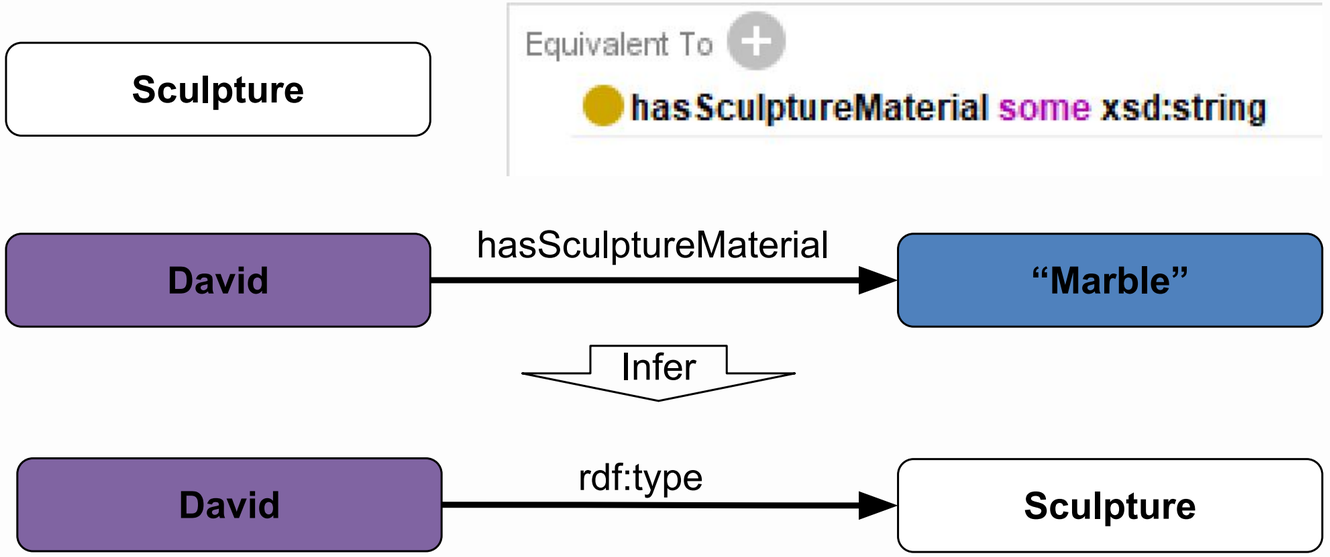
Class Definition

A sculpture has some sculpture material

RESULTS

David

Why?




```
SELECT ?x

WHERE {
    ?x a ex:Sculpture .
}
```

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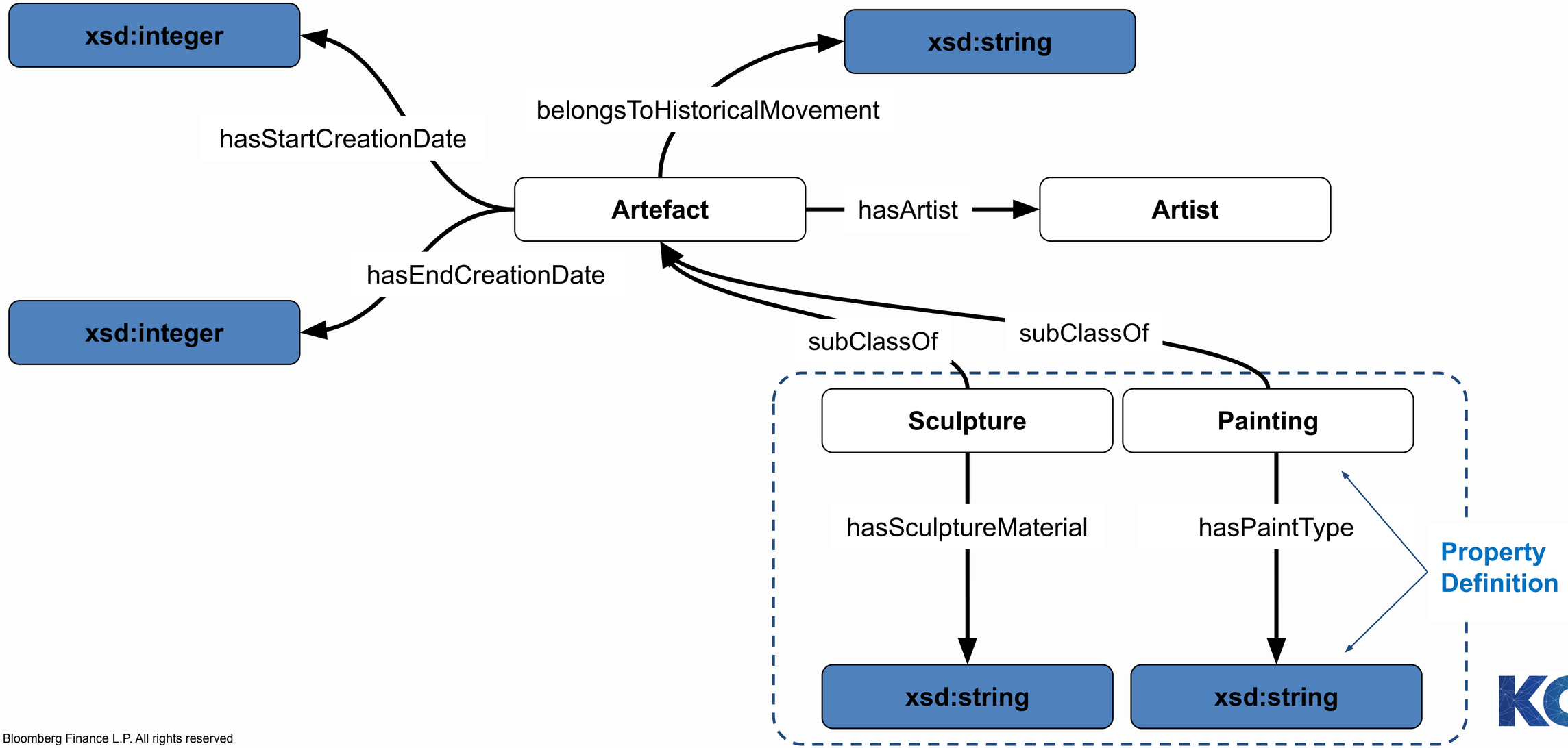
Property Definition



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Classification by Property definition

Property definition:

ex:hasPaintType owl:domain Painting

ex:hasPaintType owl:range xsd:string

At the data level:

ex mona_lisa ex:hasPaintType "Oil paint"

ex:the_last_supper ex:hasPaintType "Mural Paint"

- Add the new knowledge
  Or
- load model_07.ttl
- Run the reasoner
- Add the changes suggested by the reasoner
  Or
- upload model_07_inferred.ttl
- Run query 2.2

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Classification by Property definition

Query 2.2

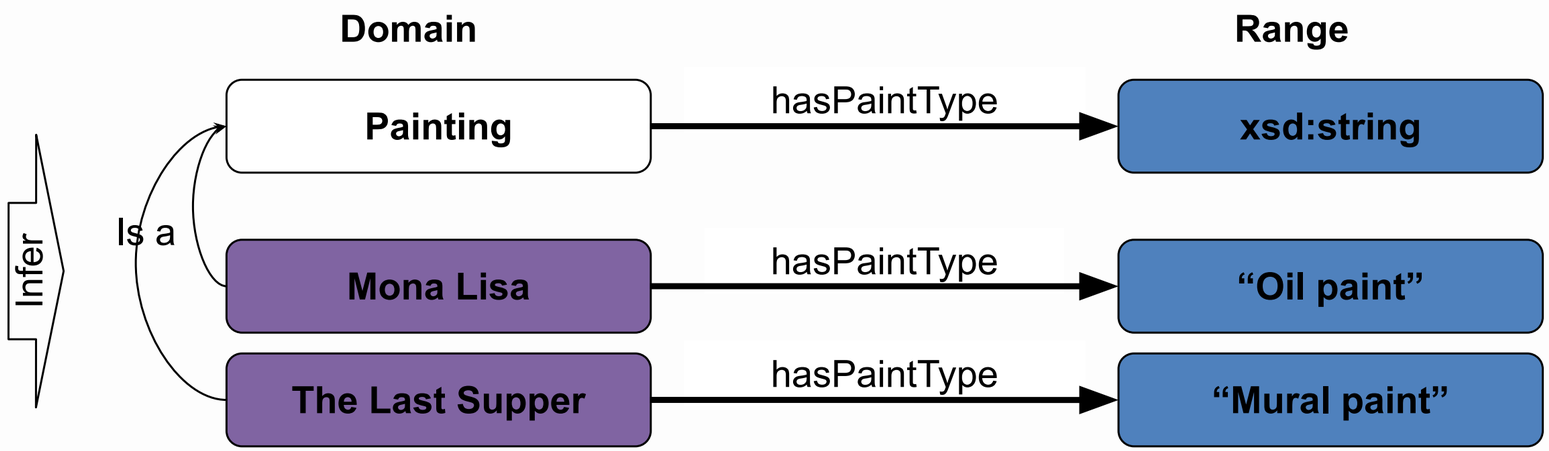
```
SELECT ?x

WHERE {
    ?x a ex:Painting .
}
```

RESULTS

Mona Lisa & Last Supper

Why? Based on the “hasPaintType” definition



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Classification Query

With all the data we have, can we query to see what are all the artefacts we have?

- load model_08.ttl

Query 2.3

```
SELECT ?x

WHERE {
    ?x a ex:Artefact .
}
```

x
ex:mona_lisa
ex:the_last_supper

We did not get "David"!

Why? David is absent because only materialised class assertions are queried (no reasoning).

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The subclass inference: Classification through inference

- Run query 2.4
  Or
- load model_09.ttl
- Run query 2.5

Query 2.4

```
SELECT DISTINCT ?x
WHERE {
   ?x rdf:type/rdfs:subClassOf*
ex:Artefact .
   }
```

Inference/reasoning logic in Query (inference on read)

Query 2.5

```
SELECT DISTINCT ?x
WHERE {
   ?x rdf:type ex:Artefact .
```

Inference/reasoning logic materialised in the knowledge graph (inference on write)

RESULTS

x
ex:mona_lisa
ex:the_last_supper
ex:david

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Inference for Classification

One of the main reasons to use OWL is its capability of making inferences

Inference can happen based on:

- Definition of the class through its properties and restrictions
- Specifying the domain and range of a property
- Through the subclass/superclass relationship

On its own, the W3C standard SHACL-Core does not support inference.

NOTE: For inference, please see SHACL Advance Features (W3C draft document).

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So how do restrictions in OWL & SHACL differ?

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OWL & SHACL restrictions compared

In OWL, the restriction purpose is Classification of resources based on the values they have:

- If something has these properties and restrictions, then it is of this class type. As we saw in the sculpture example, you have to define these restrictions in the implementation as necessary and sufficient.
- OWL is property-oriented, meaning that the definition of a class is based on its properties and the restrictions on those properties.

In SHACL, the restriction purpose is for Validation, and the logic is reverse meaning:

- If something is of a class/target type, then it has these properties
- And, if it doesn't have those properties, it will fail the validation

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Recap so far…

- Open world: missing information does not cause an error OWL
- Inference for Classification: OWL
  - Class definition
  - Property domain and range
  - Subclass relationship
- Closed world: missing information causes and error SHACL
  - Data Integrity: SHACL
  - Data Validation: SHACL

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ROUND 5: Inheritance

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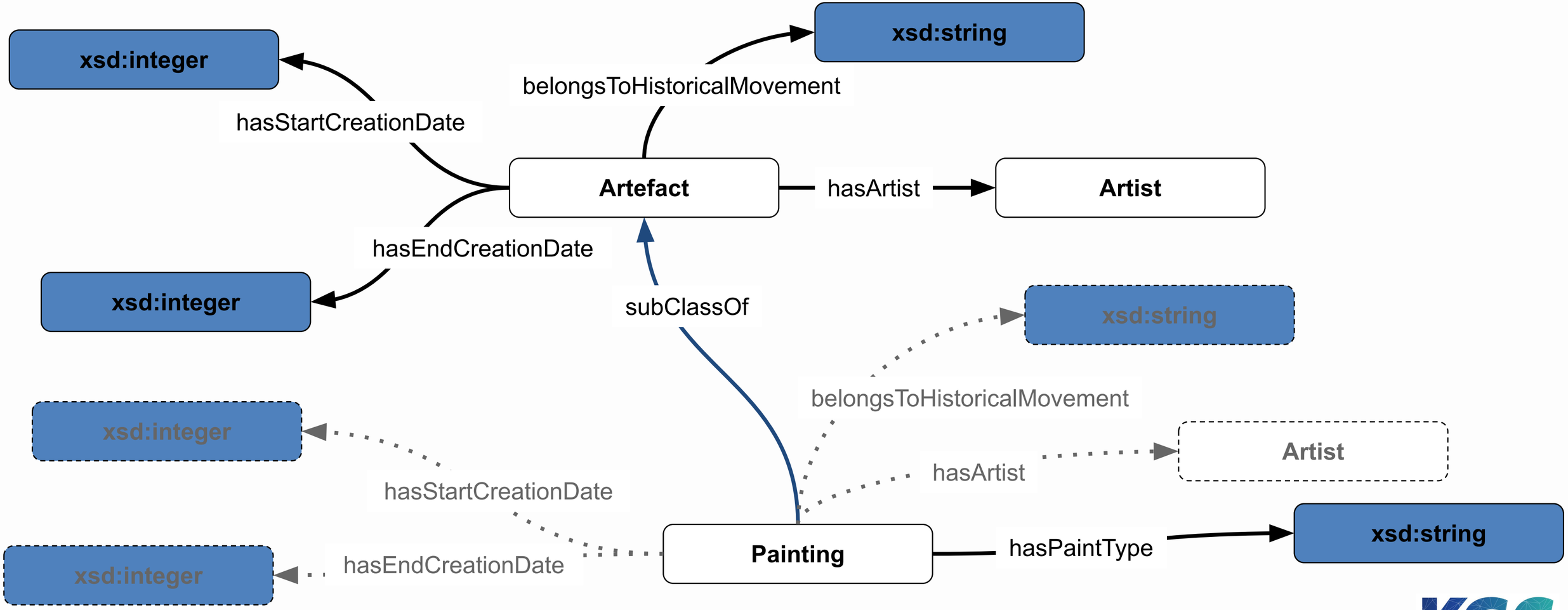
Inheritance in OWL



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Validation using inheritance in SHACL

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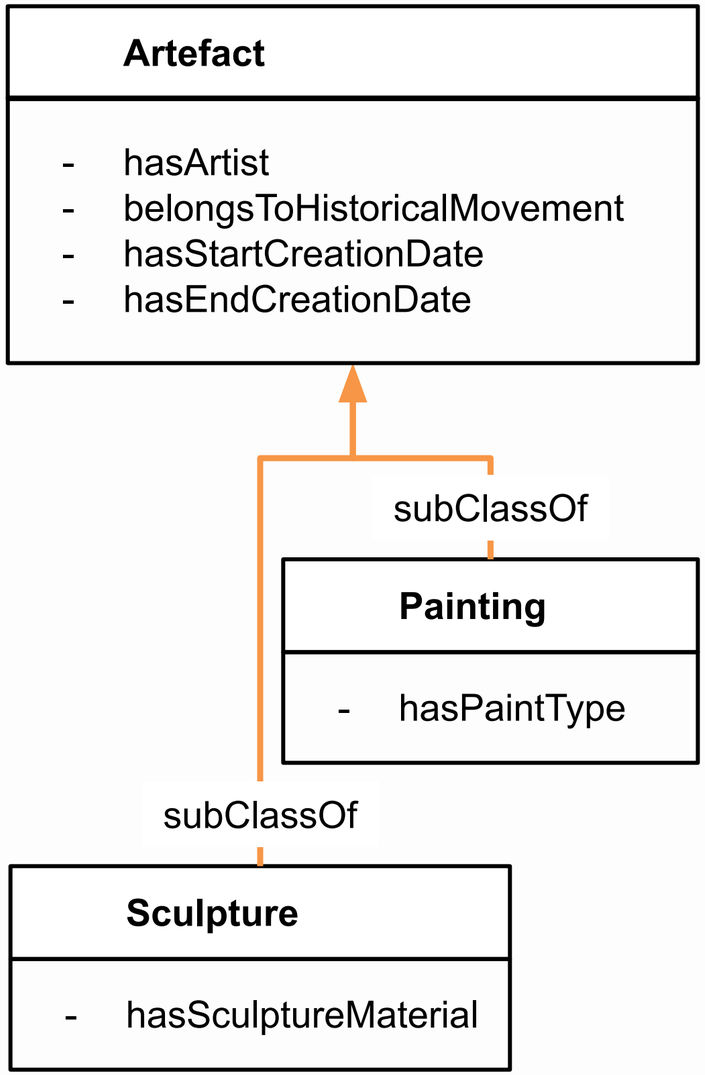
Inheritance in SHACL



A /subclassOf B

If OWL subclassing says

"Every instance of A is also an instance of B"

the SHACL sh:node says

"Every A must have data that conforms to shape B"

- Let's open Shape 3
- If I change anything in the data graph nothing happens

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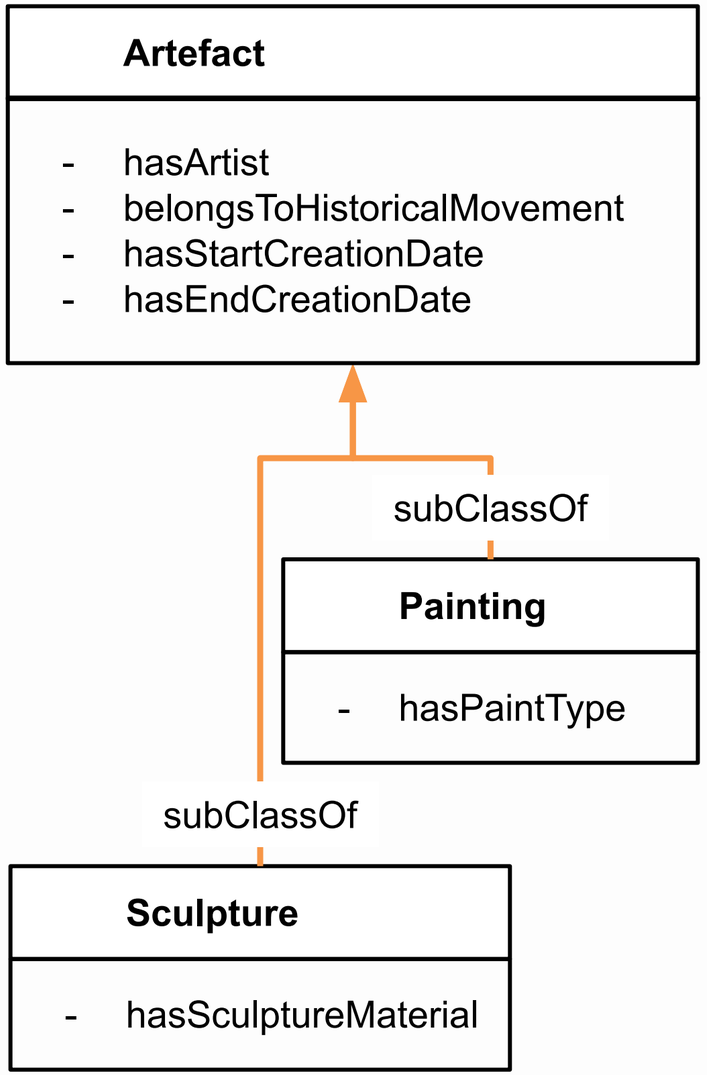
Making subClass inheritance work in SHACL



There are three ways to make it work:

1. Adding all the inference materialised in the data graph - Shape 4
2. Adding the semantic/knowledge level to the data graph - Shape 5
3. Adding sh:node on the SHACL side - Shape 6

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SHACL: Closed Shape

By default, SHACL shapes are open, which means:

- A node can have additional properties beyond those specified in the shape.
- SHACL only checks constraints on the properties you define; it ignores others.

However, SHACL also allows you to define closed shapes

- A node must only have the properties I explicitly permit here, any other properties are invalid.

```
ex:ArtefactShape a sh:NodeShape ;
  sh:targetClass ex:Artefact ;
  sh:closed true ;
  sh:ignoredProperties (rdf:type rdfs:label) ;
```

But when you close the shapes then…

- With semantic/knowledge level + sh:node | Shape 7
- With materialized data + sh:node | Shape 8
- With everything | Shape 9

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Recap so far…

- Open world: missing information does not cause an error OWL
- Inference for Classification: OWL
  - Class definition
  - Property domain and range
  - Subclass relationship
- Closed world: missing information causes and error SHACL
  - Data Integrity: SHACL
  - Data Validation: SHACL
- Inheritance: OWL
- Class composition using sh:node: SHACL

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What if we need different data restrictions for different applications?

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ROUND 6:
Multiple Shapes of Data

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One model, multiple shapes

Imagine another museum using the same ontology, but with an additional requirement: they need to ensure that the current owner or seller of an artefact is captured at the time of acquisition.

Does this mean they must create a completely new ontology? Not necessarily

- Extend the current ontology: Add properties such as isOwnedBy and classes like Person.

- Define a new shape: Define a different SHACL shape that validates their specific data needs for their use case.

```
ex:Artefact
        ex:isOwnedBy ex:Person .
```

```
sh:property [
  sh:path ex:isOwnedBy ;
  sh:class ex:Owner ;
  sh:minCount 1 ;
  sh:maxCount 1 ;
] ;
```

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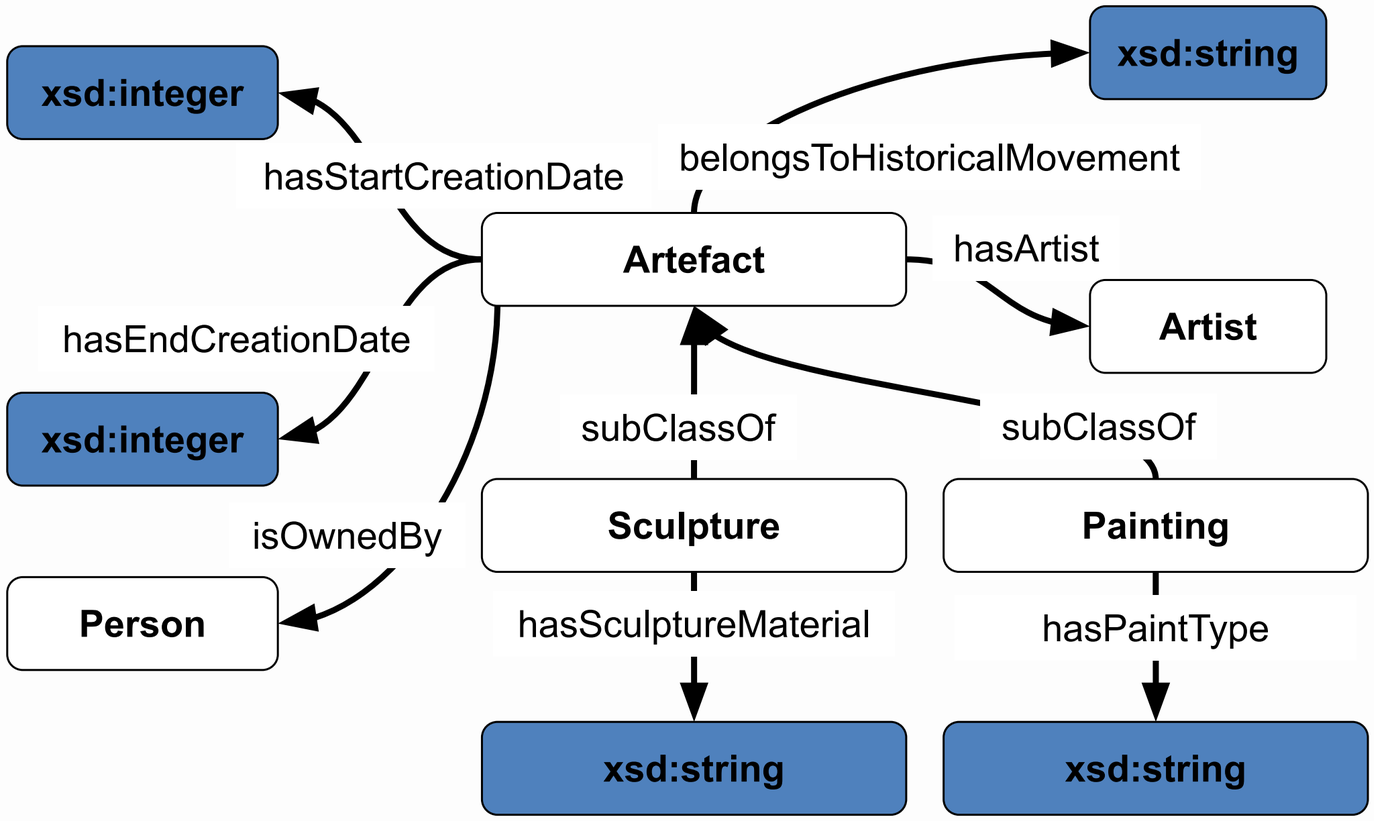
One model, multiple shapes



SHACL shape museum 1

Shape 10

SHACL shape museum 2

Shape 11

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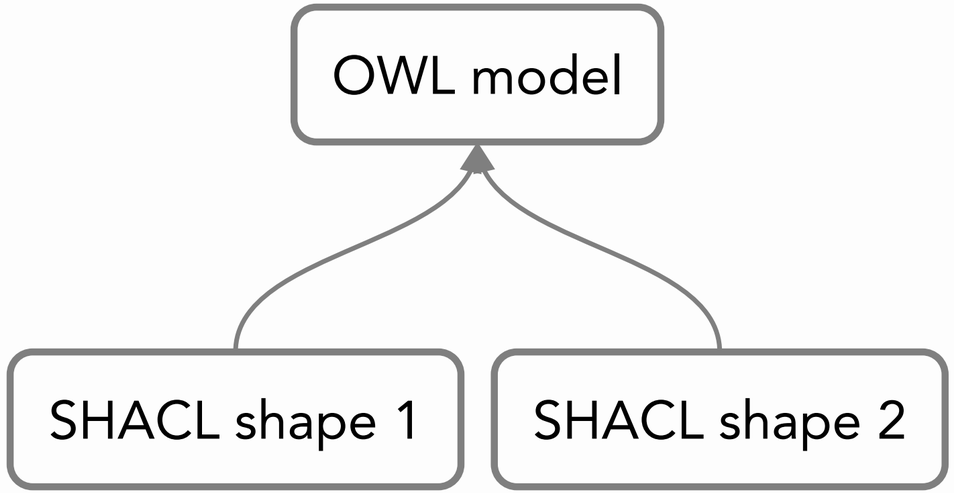
One model, multiple shapes

This ability to define multiple SHACL shapes over the same ontology supports flexibility and interoperability, allowing different institutions to share and validate data according to their own requirements while still adhering to a common semantic foundation.



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- Inference for Classification: OWL
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- Closed world: missing information causes and error SHACL
  - Data Integrity: SHACL
  - Data Validation: SHACL
- Inheritance: OWL
- Class composition using sh:node: SHACL
- Multiple shapes for the same model: OWL + SHACL

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What if want to ensure my art pieces are definitely created by an artist?

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ROUND 7: Constraints

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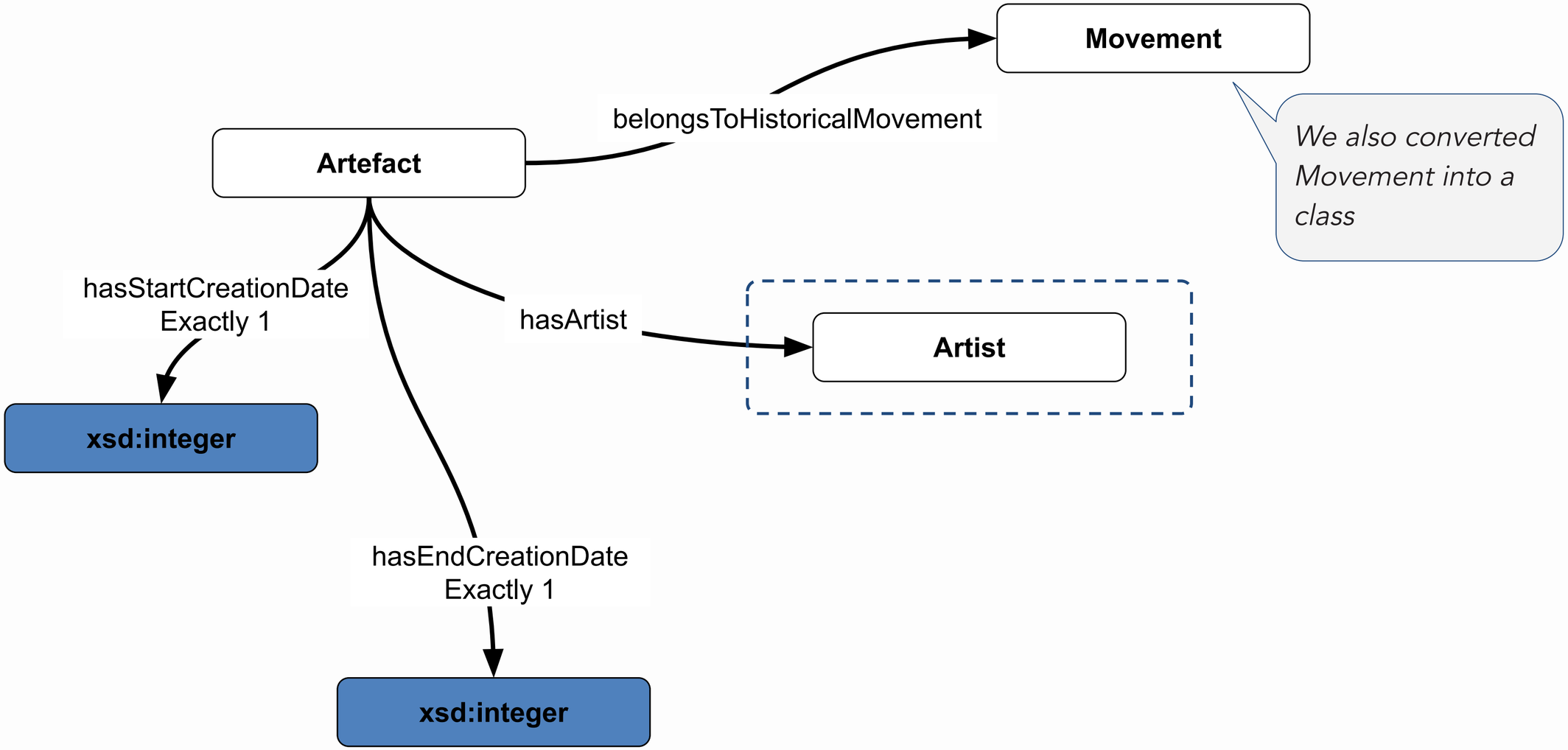
I want to ensure every art piece is only created by an artist



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Deep dive in the property definition

Can I leverage on domain and range of ex:hasArtist?

- Open model_10.ttl
- Adding wrong artist (hasArtist Cubism)

  Or
- open model_10_wrong_artist.ttl
- Run the reasoner

RESULTS

- No error even though Cubism is wrong
- Cubism classified as an Artist! (remember inference?)

WHY? 'Violating' a domain or range constraint does not necessarily mean that the ontology is inconsistent or contains errors.

So what should we do?

OWL solution: Define the two classes, Movement and Artist, as disjoint

Fun Fact:
The Weeping Woman, valued at A$1.6M in 1986. It was stolen from the National Gallery of Victoria by a group calling themselves the Australian Cultural Terrorists (ACT), who demanded funding for the arts and the creation of a $25,000 annual art prize (The Picasso Ransom) or they would destroy the painting. The government refused to negotiate, and police were informed three weeks later that *The Weeping Woman* was found, stowed in a locker, unharmed.

Source: wikipedia

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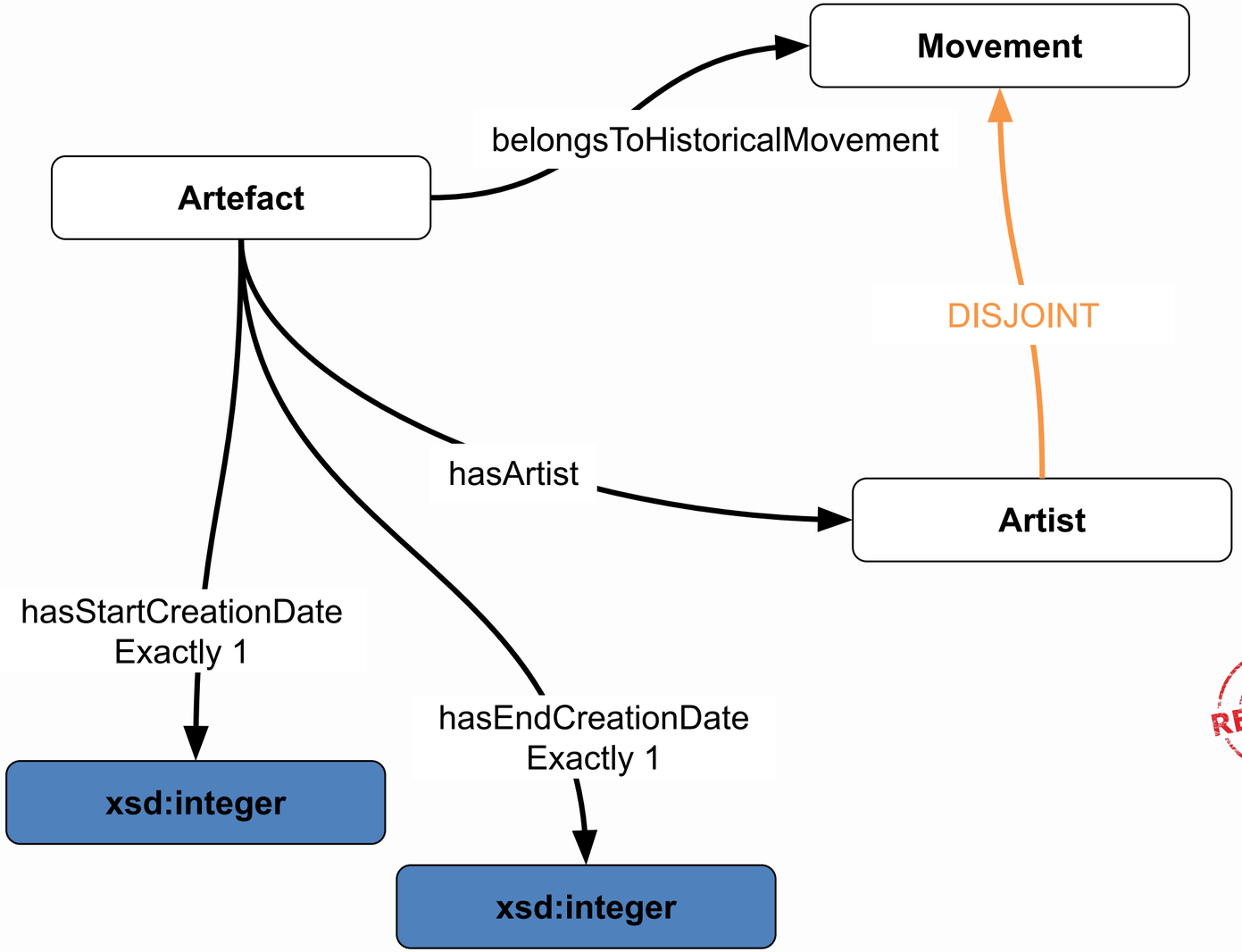
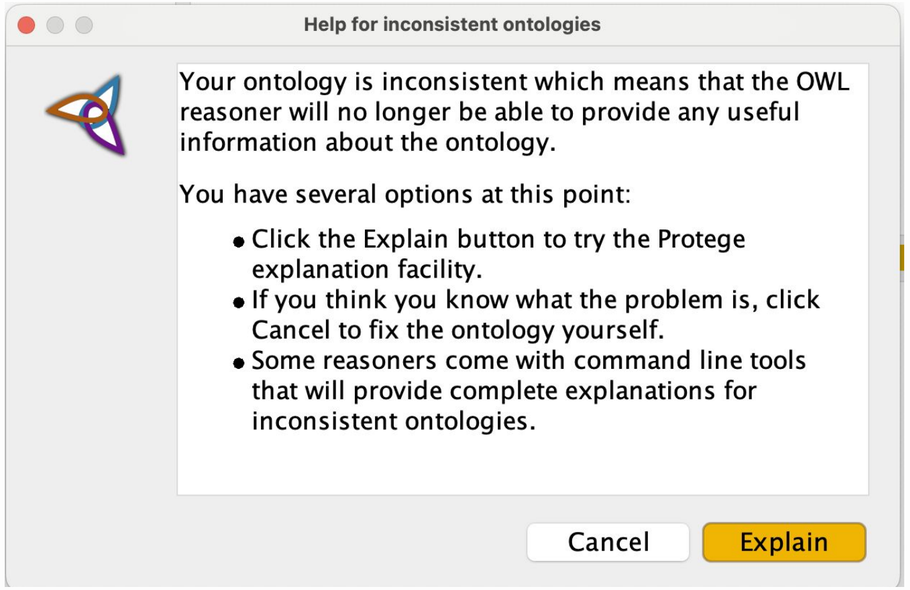
Disjoint Classes



- Make Artist and Movement disjoint
  Or
- open model_10_disjoint.ttl
- Run the reasoner



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Defining Constraints on your data with SHACL

```
ex:ArtefactShape a sh:NodeShape ;
  sh:targetClass ex:Artefact ;

  sh:property [
    sh:path ex:belongsToHistoricalMovement ;
    sh:datatype xsd:string ;
  ] ;

  sh:property [
    sh:path ex:hasArtist ;
    sh:class ex:Artist ;
  ] ;

  sh:property [
    sh:path ex:hasEndCreationDate ;
    sh:datatype xsd:integer ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
  ] ;

  sh:property [
    sh:path ex:hasStartCreationDate ;
    sh:datatype xsd:integer ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
  ] .
```

- Open Shape 12
- Change the value of The Scream ex:hasArtist to cubism

This restricts the values of ex:hasArtist to Class Artist.

RESULTS

Error! Restricting the range of ex:hasArtist to the class Artist throws an error when any other value is defined as the range in the data graph.

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SHACL Constraints

Constraints

Shape Based

- Node Shape
- Property Shape

Value Range

- sh:minExclusive
- sh:maxExclusive
- sh:minInclusive
- sh:MaxExclusive

Logical

- sh:not
- sh:and
- sh:or
- sh:xone

String Based

- sh:minLength
- sh:maxLength
- sh:pattern
- sh:languageIn
- sh:uniqueLang

Value Type

- sh:Class
- sh:datatype
- sh:nodeKind

Cardinality

- sh:minCount
- sh:maxCount

Property Pair

- sh:equals
- sh:disjoing
- sh:lessThan
- sh:lessThanOrEauals

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Inference for new (implicit) knowledge

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ROUND 8: Inference

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Inferring new Knowledge

You have the historical movement for your artefacts. Does that allow you to also know which artists belongs to what movement?

- Open model_11.ttl
- Run the query 3.1

RESULTS Nothing!

```
SELECT DISTINCT ?Artist ?Movement
WHERE {
   ?Artist a ex:Artist ;
     ex:belongsToHistoricalMovement ?Movement .
}
```

Possible solution: Property Chain

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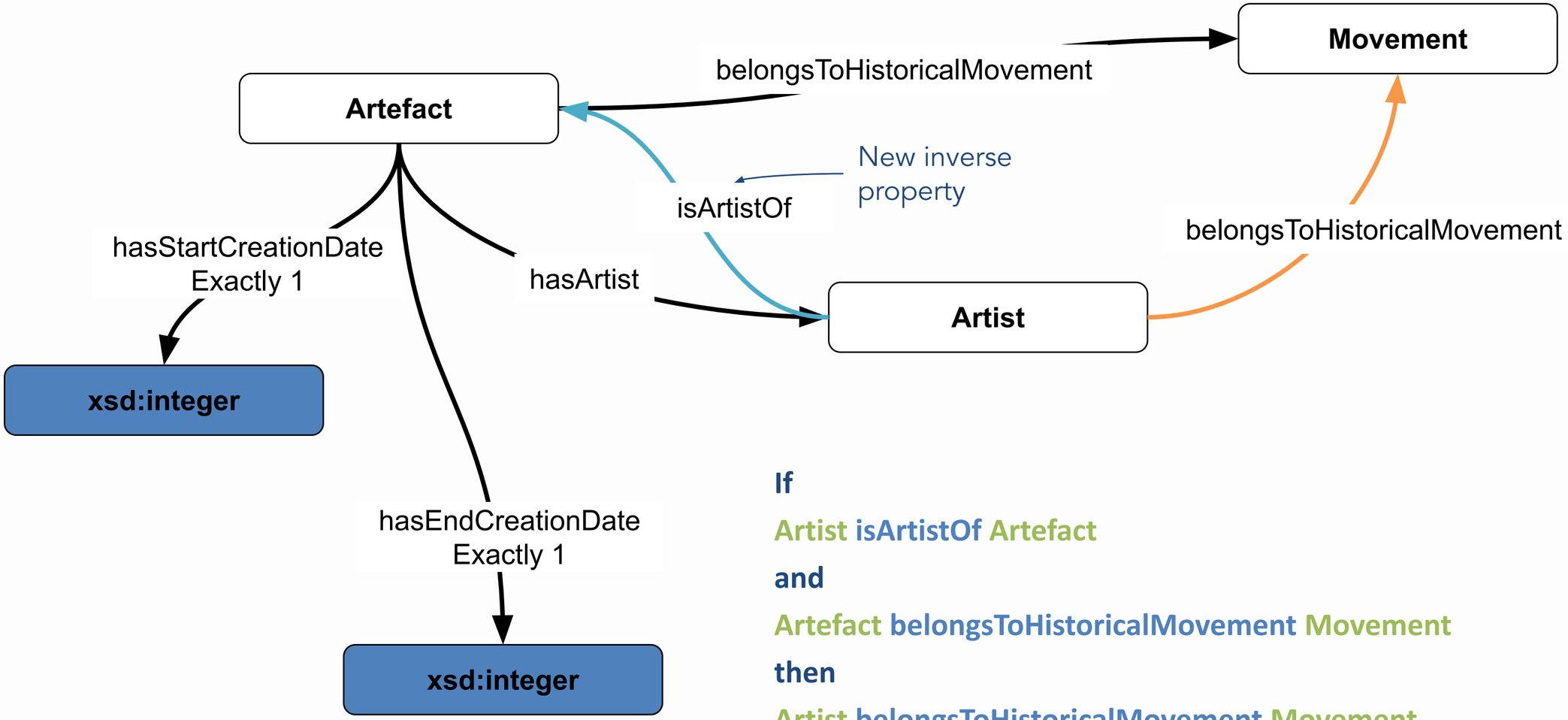
The model extended even more!



If

Artist isArtistOf Artefact

and

Artefact belongsToHistoricalMovement Movement

then

Artist belongsToHistoricalMovement Movement

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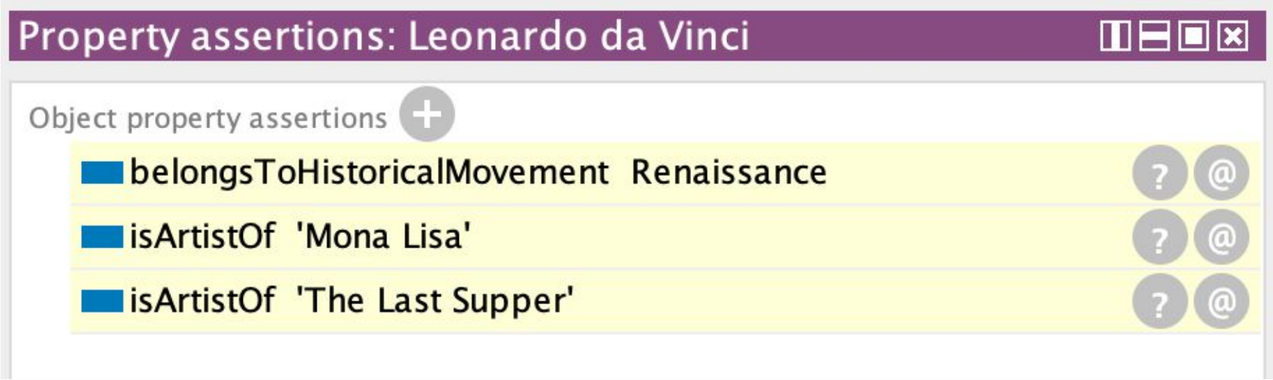
Inverse properties in OWL

- First, add isArtistOf inverse property
- Implement the property chain
  Or
- load model_12.ttl
- Run the reasoner

RESULTS Reasoner infers the implicit knowledge



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Property chain in OWL (owl:propertyChainAxiom)

- Save the new inferred data
  Or
- upload model_12_inferred.ttl
- Then run the query 3.1 again

```
SELECT DISTINCT ?Artist ?Movement
WHERE {
    ?Artist a ex:Artist ;
      ex:belongsToHistoricalMovement ?Movement .
}
```

RESULTS

Artist	Movement
Leonardo da Vinci	Renaissance
Edvard Munch	Expressionism
Pablo Picasso	Cubism
Sandro Botticelli	Renaissance
Michelangelo	Renaissance

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Inferring new knowledge

Considering that you know the museum in which the artpiece is exhibited, can you answer questions like:

Give me all the pieces that are located in a country (i.e., Italy)?

In order to answer to questions like this, we need to add further knowledge into our KG:

If

Artefact *isExhibitedIn* Premises

and

Premises *isLocatedIn* City

then

Artefact *isLocatedIn* City

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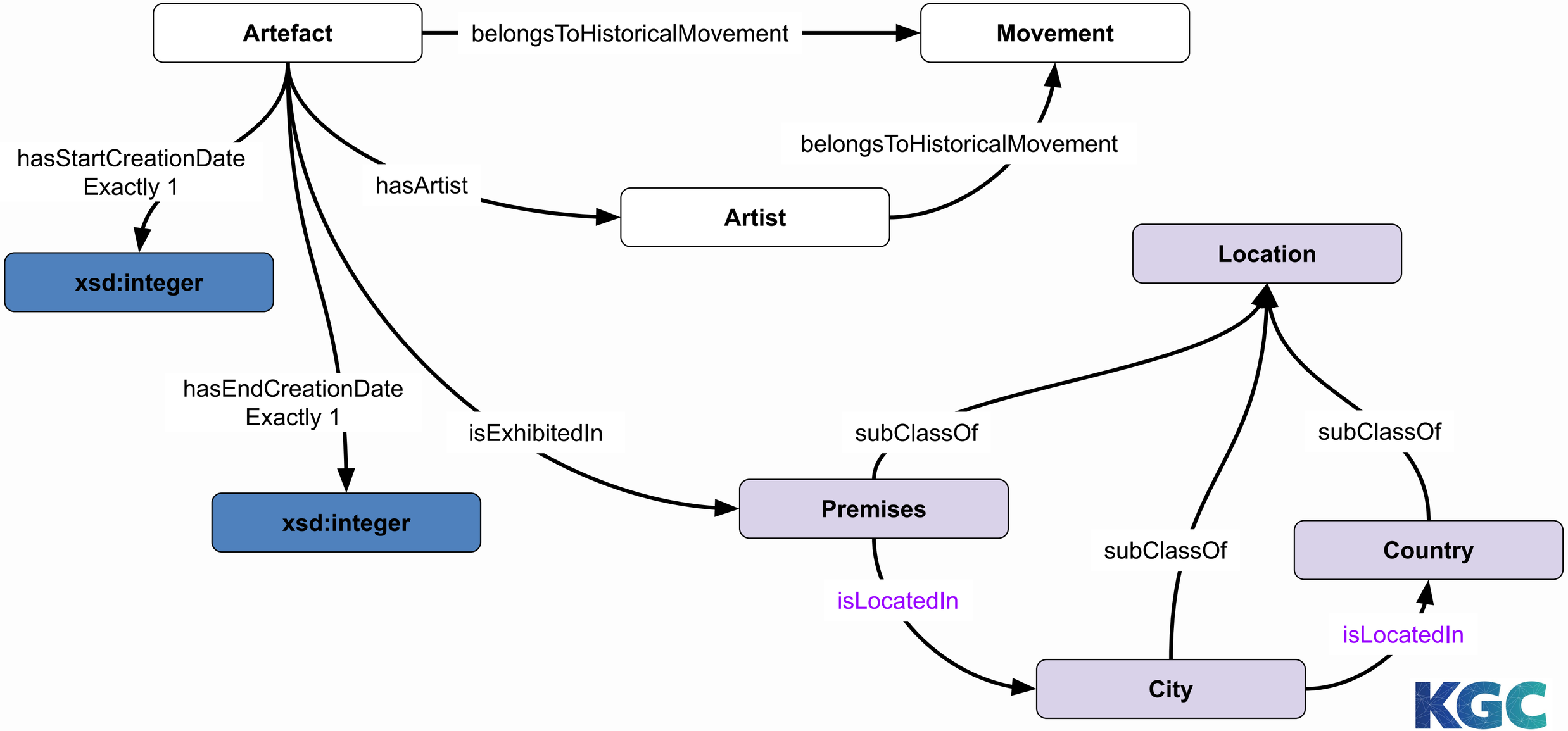
The model extended even more!



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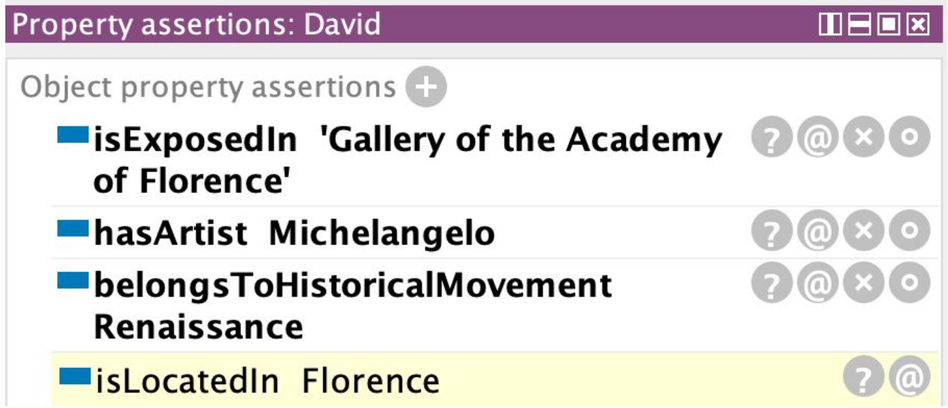
Inferring new knowledge

- Add new classes and the property chain
  Or
- Load model_13.ttl
- Run reasoner

Reasoner infers the implicit Knowledge based on property chain definition



- Save the new inferred data
  Or
- Upload model_13_inferred.ttl
- Run query 3.2

```
SELECT DISTINCT ?Artefact
WHERE {
?Artefact a ex:Artefact ;
ex:isLocatedIn ex:italy .
}
```

Nothing!
Why? We are still missing the country

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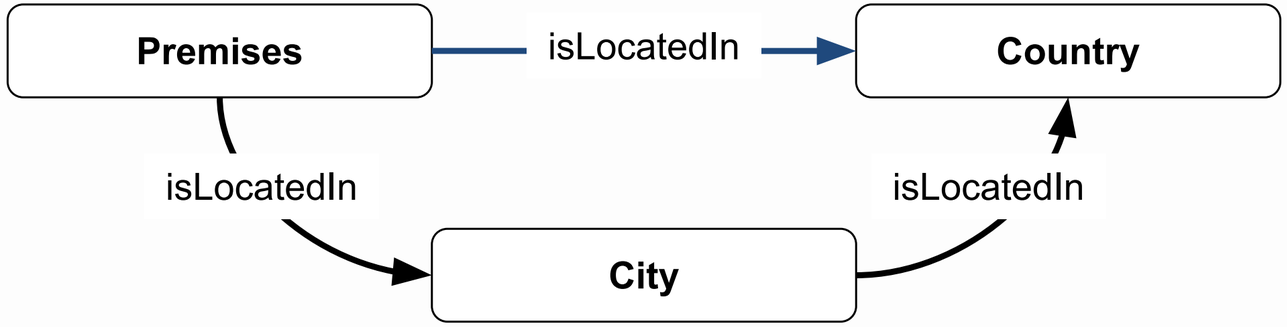
Transitivity in properties for inference of new knowledge

How can we ensure that if something is located in a city then that something is located in the country of that city?

Transitivity automatically infers that if a property links A → B and B → C, then A → C for that same property so we can say:



owl:propertyChainAxiom allows defining a new property that results from a sequence of different properties (e.g., A → B via isExhibitedIn and B → C via isLocatedIn ⇒ A → C via isLocatedIn).

isLocatedIn is transitive property. Then Artefact isLocatedIn Country.

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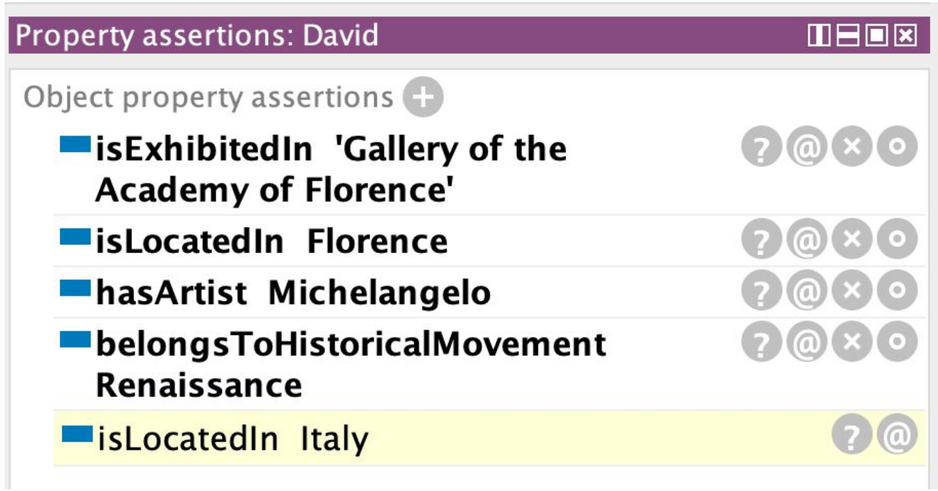
Transitivity in properties for inference of new knowledge

- Make isLocatedIn transitive
  Or
- Open model_14.ttl
- Run reasoner

Reasoner infers the implicit Knowledge based on property chain definition and transitivity



- Save the new inferred data
  Or
- upload model_14_inferred.ttl
- Run query 3.2 again

You can now see all artefacts located in Italy

```
SELECT DISTINCT ?Artefact
WHERE {
?Artefact a ex:Artefact ;
ex:isLocatedIn ex:italy .
}
```

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- Class composition using sh:node: SHACL
- Inference for new/implicit knowledge: OWL
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The Material Example

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ROUND 9:
Range Restrictions

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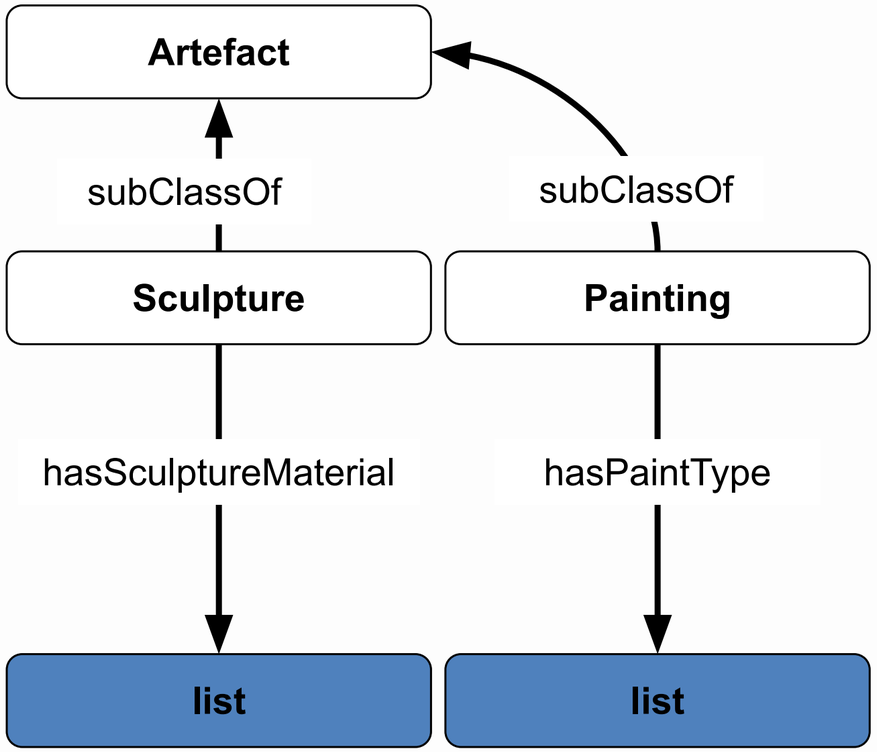
The Material Example

What are the materials used to make this artefact?
You also want to ensure that the range of allowed materials is previously captured.

Let's start with the model: model_15.ttl

Choice 1: use OWL dataproperty and a list as the range.



Challenge:

- What if you want to say something more about each material? Like its origin or consistency?

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The Material Example

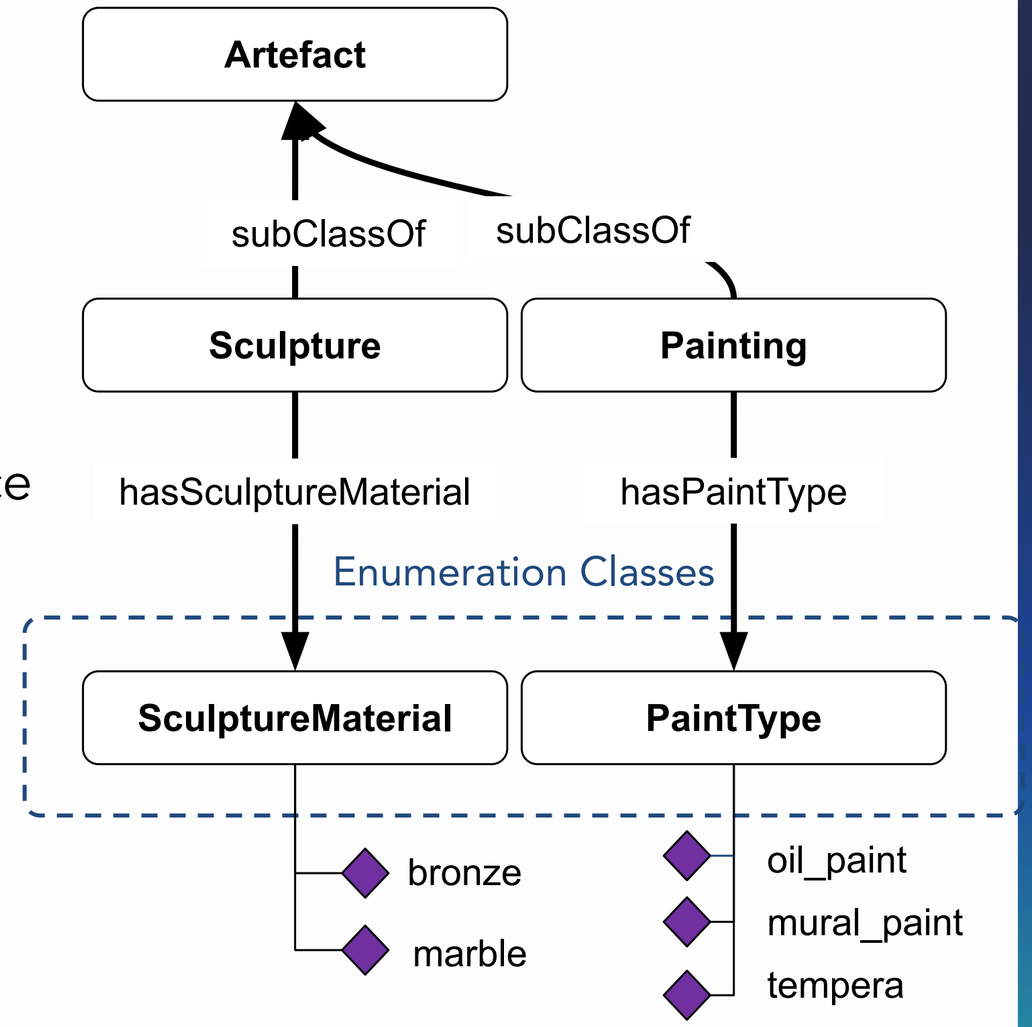
What are the materials used to make this artefact?

Choice 2: Use OWL Enumeration class

- Load model_16.ttl
- Add the following wrong data
  ex:weapingWoman ex:hasPaintType ex:florence
- Run reasoner

No Error!
Reasoner infers that florence is a member of *PaintType* class

WHY? Because of OWL's OWA



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The Material Example

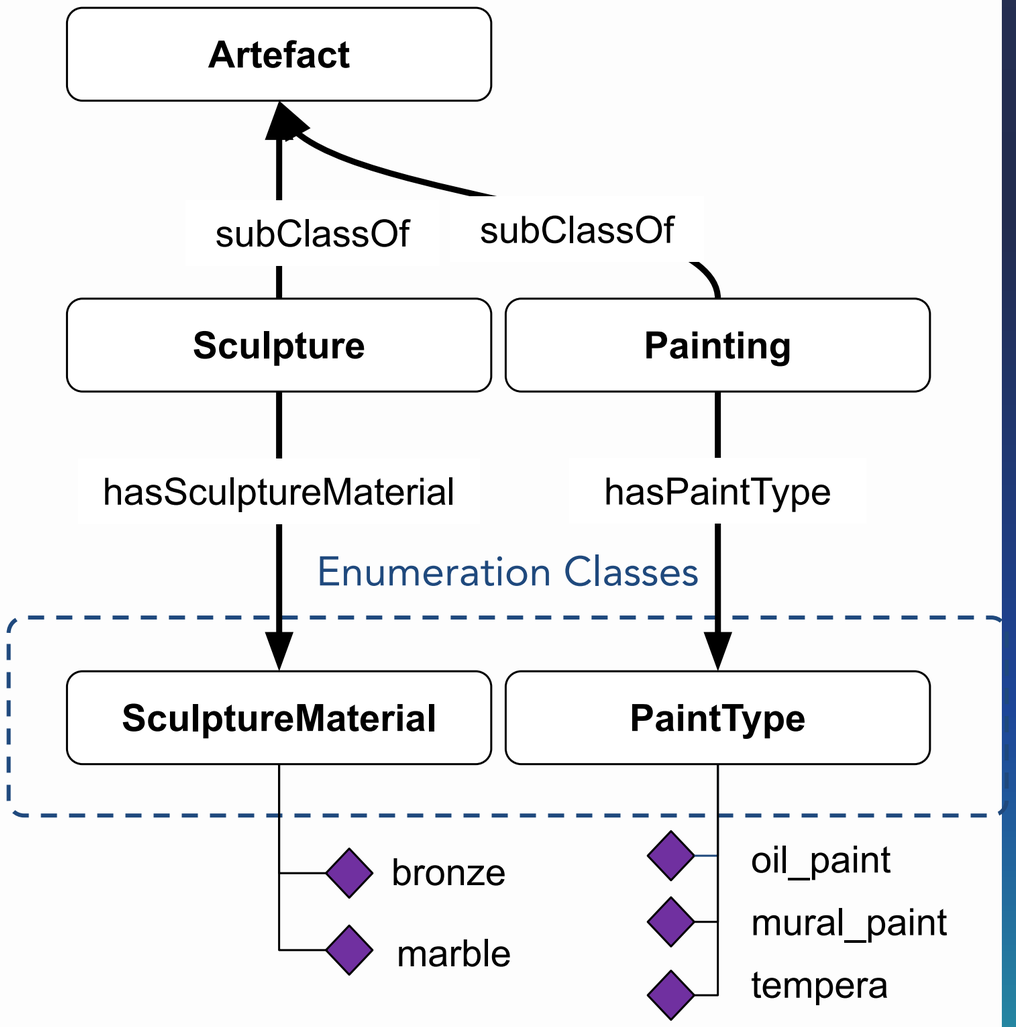
What are the materials used to make this artefact?

Choice 2: Use OWL Enumeration class

- What if we limit the range of *hasPaintType* to <u>only one</u> *PaintType*?
- Making the property hasPaintType Functional

No Error!
Reasoner infers that the instances are the same!

WHY? OWL does not support UNA



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Unique Name Assumption

In logics with Unique Name Assumption (UNA), different names always refer to different entities in the world.

OWL does NOT support UNA

Consequences:

- Different entities have to be declared to be different (otherwise, they are potentially identical) - owl:differentFrom, owl:disjointWith
- Identical entities also have to be declared to be identical (otherwise, they are potentially different) - owl:sameAs, owl:equivalentClass

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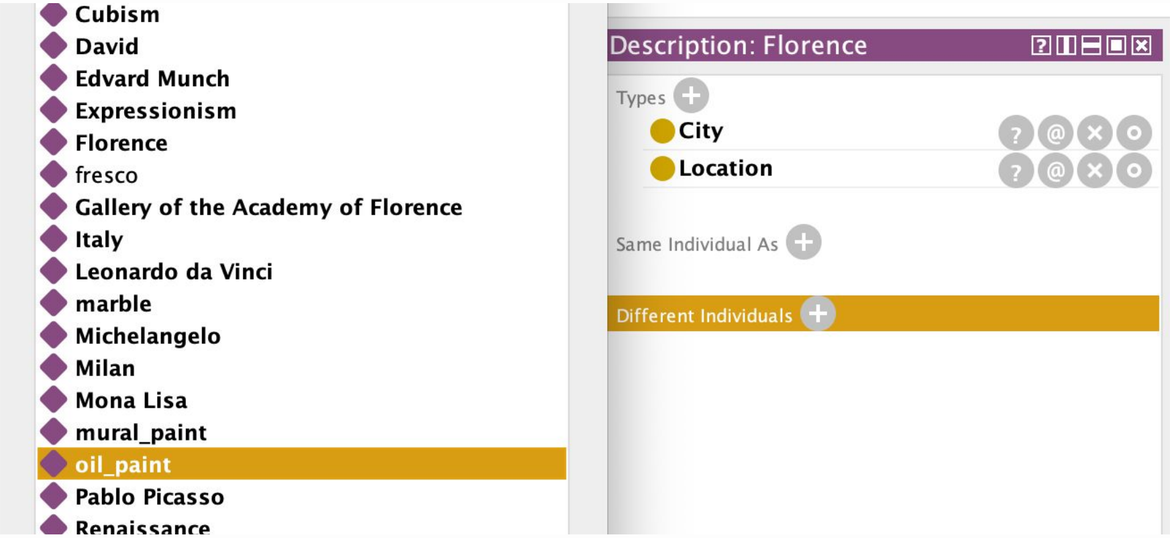
Can we still solve this with OWL?

Yes, but…

The property hasPaintType needs to be Functional

The instances need to all be defined as owl:differentFrom



Too much work!

There's a better solution :)

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The Material Example

What are the materials used to make this artefact?

Choice 3:

- Use OWL enumeration classes & SHACL shapes to restrict the range of artefacts Shape 13

Pros:

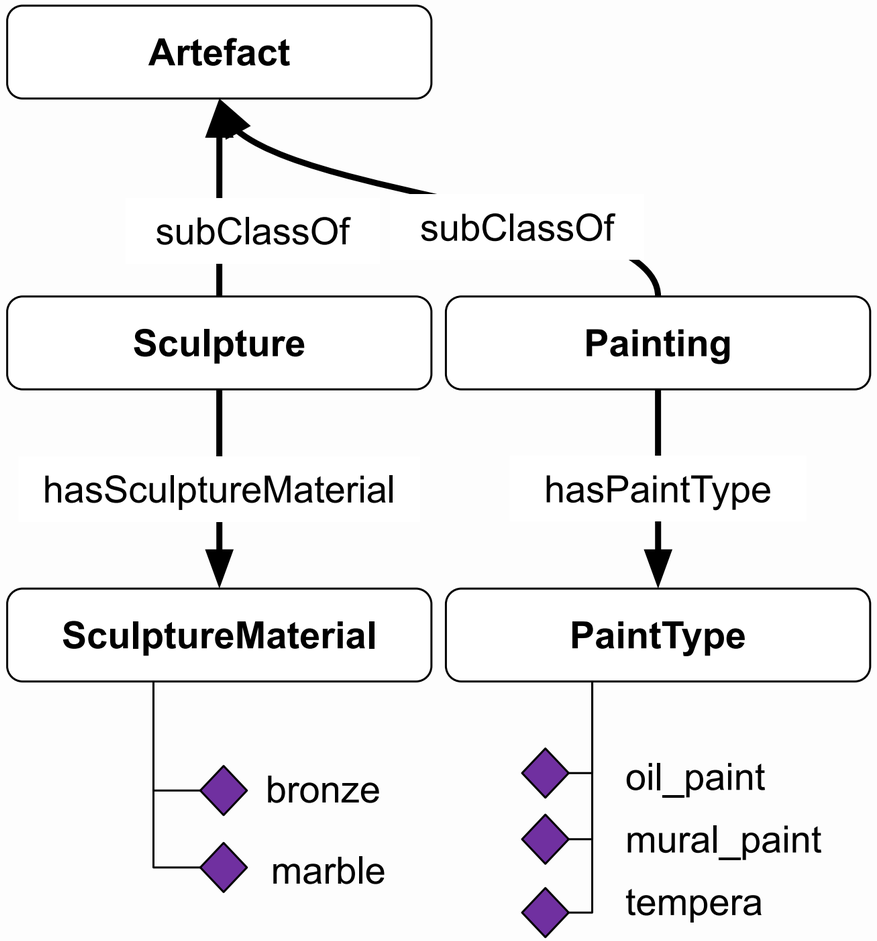
- No diff definition
- No wrong class inference
- Easy expansion of material types
- Easy addition of information about material

```
ex:PaintingShape a sh:NodeShape ;
  sh:targetClass ex:Painting ;

  sh:property [
    sh:path ex:hasPaintType ;
    sh:class ex:PaintType ;
    sh:in (ex:oil_paint ex:mural_paint ex:tempera) ;
  ] .

ex:SculptureShape a sh:NodeShape ;
  sh:targetClass ex:Sculpture ;

  sh:property [
    sh:path ex:hasSculptureMaterial ;
    sh:class ex:SculptureMaterial ;
    sh:in (ex:marble ex:bronze) ;
  ] .
```



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Full Recap

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  - Class definition OWL
  - Property domain and range OWL
  - Subclass relationship OWL
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  - Data Validation: SHACL
  - Restrictions/Constraints for validation: SHACL
- Inheritance: OWL
- Class composition using sh:node: SHACL
- Multiple shapes for the same model: OWL + SHACL
- Mix of Inference and Validation: OWL + SHACL
- Inference for new/implicit knowledge: OWL

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Bonus ROUND : LLMs to generate OWL and SHACL

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Where LLMs help in the OWL/SHACL workflow

1. Requirements → modeling candidates

Extract candidate classes, properties, competency questions, constraints, synonyms, and edge cases from prose.

2. Draft artifacts

Produce first-pass OWL/Turtle, SHACL node/property shapes, examples, SPARQL checks, and documentation.

3. Explain & translate

Turn terse validation or reasoning results into stakeholder-friendly explanations and remediation guidance.

4. Review support

Suggest alternative patterns, test cases, competency-question coverage gaps, and potential inconsistencies.

5. Migration & alignment

Map legacy schemas to target vocabularies, propose URI naming, align terms, and draft mapping notes.

6. Learning scaffold

Help practitioners understand why OWL and SHACL differ through examples, counterexamples, and walkthroughs.

Strongest use case: accelerate iteration between domain experts, ontology engineers, and data stewards.

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Benefits and failure modes of LLM-generated SHACL and OWL

PROS

Speed

Rapidly produces initial OWL, SHACL, examples, and documentation from natural-language requirements.

Coverage prompts

Can propose competency questions, constraints, negative examples, and test cases practitioners may miss.

Accessibility

Helps non-specialists understand validation reports and semantic-modeling choices.

Pattern exploration

Can compare modeling alternatives and generate variants for expert discussion.

RISKS

Hallucinated terms

May invent classes, properties, namespaces, axioms, or business rules that look plausible.

Semantic drift

May confuse OWL meaning with SHACL validation, or produce artifacts that are syntactically valid but conceptually wrong.

Hidden assumptions

May omit scope, cardinality rationale, target selection, closed/open-world implications, or inference dependencies.

No quality guarantee

Requires parsers, reasoners, SHACL engines, tests, and expert review before production use.

KGC

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Recommended governance pattern: human-in-the-loop by design

1 Scope

Business purpose
CQs
non-goals

2 Prompt

Known patterns
existing
vocabularies
examples

3 Generate

OWL / SHACL
docs
test data

4 Machine check

Parse
reason
validate

5 Expert review

Semantics
intent
edge cases

6 Publish

Version
trace
monitor

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FINAL ROUND:
Just a fun fact

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The Latest Louvre Heist!

Speed-Run Art Theft

The crew pulled off the whole job in **7 minutes** — with only **4 minutes inside** The Louvre.

Bob the Builder: Heist Edition

They showed up in high-vis construction vests, carrying power tools, and scaled the museum using a **moving lift** before zipping away on **scooters**.

Which the moving lift company then used to advertise!

Zero Bubble-Wrap

They stole **French crown jewels** from Empress Eugénie and other royals. One piece was even **dropped and damaged** during the escape.

Created using ChatGPT + Gemini

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And the winner is……

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WINNER: YOU

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- Identify and understand the true problem you are trying to solve
- Understand the strengths and weaknesses of each technology and why and what it's built for
- Choose the right tooling and understand the strengths of your tool
- Based on the above KNOWLEDGE infer the RIGHT CHOICE!

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Thank You!

Let’s discuss more…
Connect with us on LinkedIn

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KGC The Knowledge Graph Conference

#KGC2026

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How do they organize their data?

Exhibited In

Related artefacts

Artefact

Owner

Movements

Artist

Exhibition

Culture

Material

Collection

Creation Date

Location

Provenance

Visitor

Period/Era

Influence

Concept

Technique

Documentation

KGC

What is SHACL?

SHACL shapes are required to validate RDF data against specific structural and value constraints.

We need to define the exact shape of our data, including the required properties and any constraints on their values.

In SHACL, a shape is composed of targets and constraints.

- Targets identify the nodes in the data graph that the shape applies to.
- Constraints specify the rules that those nodes must satisfy in order to be considered valid.

Declare constraints directly on a node

Targets

Node shape

```
ex:ArtefactShape a sh:NodeShape ;
  sh:targetClass ex:Artefact ;

  sh:property [
    sh:path ex:hasEndCreationDate ;
    sh:datatype xsd:integer ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
  ] ;
```

Property shape

Constraints

Declare constraints on the property that is connected to the node through a path (the sequence of edges through which a property is connected to a node)

Where LLMs help in the OWL / SHACL workflow

Use them to reduce blank-page effort and improve communication — not to bypass modeling decisions.

1. Requirements → modeling candidates

Extract candidate classes, properties, competency questions, constraints, synonyms, and edge cases from prose.

2. Draft artifacts

Produce first-pass OWL/Turtle, SHACL node/property shapes, examples, SPARQL checks, and documentation.

3. Explain & translate

Turn terse validation or reasoning results into stakeholder-friendly explanations and remediation guidance.

4. Review support

Suggest alternative patterns, test cases, competency-question coverage gaps, and potential inconsistencies.

5. Migration & alignment

Map legacy schemas to target vocabularies, propose URI naming, align terms, and draft mapping notes.

6. Learning scaffold

Help practitioners understand why OWL and SHACL differ through examples, counterexamples, and walkthroughs.

Strongest use case: accelerate iteration between domain experts, ontology engineers, and data stewards.

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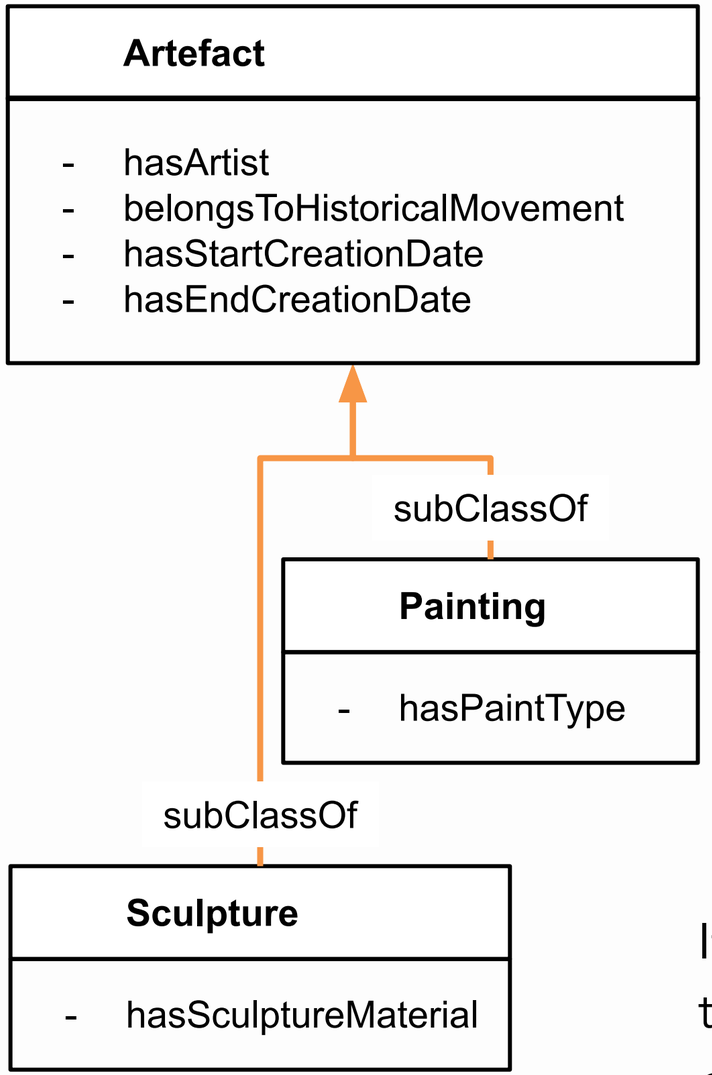
Making subClass inheritance work in SHACL



- Run Shape 3
- The data graph conforms to the shape, hence, no error
- If we change one of the date from integer to string

There are three ways to make it work:

1. Adding all the inference materialised in the data graph - Shape 4
2. Adding the semantic/knowledge level to the data graph - Shape 5
3. Adding sh:node on the SHACL side - Shape 6

If OWL subclassing says "*Every instance of A is also an instance of B*", the SHACL sh:node says "*Every A must have data that conforms to shape B*"

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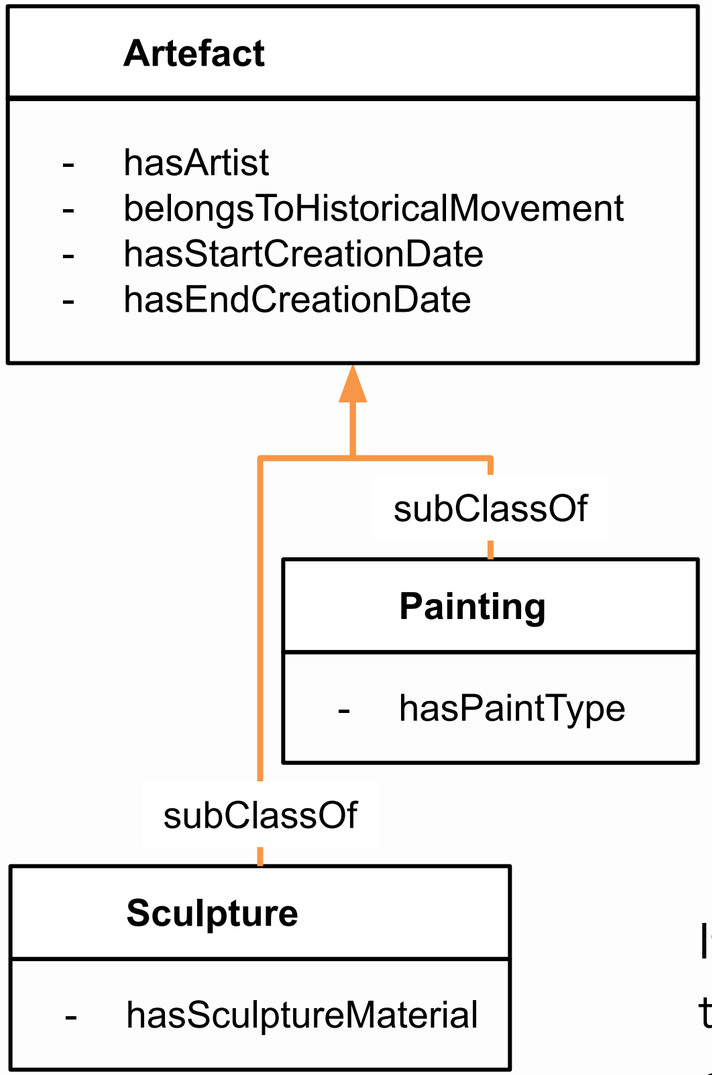
Inheritance in SHACL



- Run Shape 3
- The data graph conforms to the shape, hence, no error

- If we change one of the dates from integer to string

Why? Because SHACL shapes are open and it only checks for the properties mentioned in the shape and doesn't care about the rest.

If OWL subclassing says "*Every instance of A is also an instance of B,*" the SHACL sh:node says "*Every A must have data that conforms to shape B.*"

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LLMs, SHACL and Owl

- We can use LLMs to generate these artifacts.
- We still need to know where to use which and how they work (and also understand the purpose they serve) this will allow you to ensure you are still using the right artifact for the purpose.
- …..
- An LLM can generate syntax; it cannot reliably decide the intended semantics without expert guidance.

Using LLMs to Generate OWL & SHACL Artifacts

Acceleration is real. Governance and foundational knowledge remain non-negotiable.

Recommended framing: LLMs are drafting assistants, not semantic authorities.